

MATRICIAL GAUSSIAN QUADRATURE RULES: SINGULAR CASE

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ABSTRACT. Let L be a linear operator on univariate polynomials of bounded degree taking values in real symmetric matrices, whose moment matrix is positive semidefinite. Assume that L admits a positive matrix-valued representing measure μ . Any finitely atomic representing measure with the smallest sum of the ranks of the matricial masses is called minimal. In this paper, we characterize the existence of a minimal representing measure that contains a prescribed atom with a prescribed rank of the corresponding mass, thereby generalizing our recent result [ZZ26], which addresses the same problem in the case where the moment matrix is positive definite. As a corollary, we obtain a constructive, linear-algebraic proof of the strong truncated Hamburger matrix moment problem.

1. INTRODUCTION

In this paper we study matricial Gaussian quadrature rules for a linear operator L on univariate polynomials of bounded degree with values in real symmetric matrices. We work in the setting where the truncated moment matrix (see (2.2)) is positive semidefinite and L admits a positive matrix-valued representing measure. More precisely, for a fixed real number t and an integer $m \in \mathbb{N} \cup \{0\}$, we characterize the conditions under which there exists a minimal representing measure for L that contains t in its support and whose corresponding mass has rank m . Our results extend the positive definite case treated in [ZZ26] to the singular setting. These results will be important in solving the truncated univariate matrix rational moment problem, in analogy with the scalar case studied in [NZ25]. We present the positive semidefinite case separately, since—unlike in the positive definite situation—the moment matrix exhibits nontrivial column relations that, due to its recursively generated structure, propagate to neighboring block columns on the right. This propagation significantly increases the technical complexity of both the construction and the proofs.

Let $i, j \in \mathbb{Z}$. We write $[i; j] := \{i, i+1, \dots, j\}$ if $i \leq j$, and $[i; j] := \emptyset$ if $i > j$. For $j \in \mathbb{Z}$, we write $[j] := [1; j]$ for short. In particular, $[j] = \emptyset$ for $j \leq 0$. Let $k \in \mathbb{N} \cup \{0\}$ and $p \in \mathbb{N}$. We denote by $\mathbb{R}[x]_{\leq k}$ the vector space of univariate polynomials of degree at most k and by $\mathbb{S}_p(\mathbb{R})$ the set of real symmetric matrices of size $p \times p$. For a given linear operator

$$(1.1) \quad L : \mathbb{R}[x]_{\leq 2n} \rightarrow \mathbb{S}_p(\mathbb{R})$$

denote its **matricial moments** by $S_i := L(x^i)$ for $i \in [0; 2n]$. Assume that L admits a positive $\mathbb{S}_p(\mathbb{R})$ -valued measure μ (see Subsection 2.1), i.e.,

$$(1.2) \quad L(p) = \int_{\mathbb{R}} p \, d\mu \quad \text{for every } p \in \mathbb{R}[x]_{\leq 2n}.$$

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Every measure μ satisfying (1.2) is called a **representing measure for L** . By [BW11, Theorem 2.7.6], whenever L has a representing measure, there is one of the form $\mu = \sum_{j=1}^{\ell} \delta_{x_j} A_j$, where each $0 \neq A_j \in \mathbb{S}_p(\mathbb{R})$ is positive semidefinite, $\sum_{j=1}^{\ell} \text{rank } A_j = \text{rank } M(n)$ with $M(n)$ being a truncated moment matrix of $(L(x^i))_{i \in [0; 2n]}$ (see (2.2)) and δ_{x_j} stands for the Dirac measure supported in x_j . Such a measure is called **minimal** because the total sum of the ranks of its matricial masses is minimal among all representing measures for L . In this case (1.2) is equal to

$$(1.3) \quad L(p) = \sum_{j=1}^{\ell} p(x_j) A_j \quad \text{for every } p \in \mathbb{R}[x]_{\leq 2n},$$

and (1.3) is called a **matricial Gaussian quadrature rule for L** . The points x_j are called **atoms** of the measure μ . If $x_1, \dots, x_{\ell}$ are pairwise distinct, then for each j , the matrix $A_j = \mu(\{x_j\})$ is called the **mass** of μ at x_j and its rank is called the **multiplicity** of x_j in μ , which we denote by $\text{mult}_{\mu} x_j$. If x is not an atom of μ , then $\text{mult}_{\mu} x := 0$.

The central problem of this paper is the following:

Problem. Let L be as in (1.1) and assume that it admits a positive matrix-valued representing measure. Given $t \in \mathbb{R}$ and $m \in \mathbb{N} \cup \{0\}$, characterize the existence of a minimal representing measure μ for L such that $\text{mult}_{\mu} t = m$.

The main result of the paper is the solution to the Problem above.

Theorem 1.1. Let $n, p \in \mathbb{N}$ and $L : \mathbb{R}[x]_{\leq 2n} \rightarrow \mathbb{S}_p(\mathbb{R})$ be a linear operator, which admits a representing measure. Fix $t \in \mathbb{R}$ and $m \in \mathbb{N} \cup \{0\}$. Let $T_i := L((x - t)^i)$ for $i \in [0; 2n]$ and

$$\mathcal{M}_1 := (T_{i+j-2})_{i,j=1}^{n+1}, \quad \mathcal{M}_2 := (T_{i+j-2})_{\substack{i=1,\dots,n, \\ j=1,\dots,n+2}}.$$

Denote by $\mathbf{y}_j$ the j -th column of $\mathcal{M}_1$ and by $\mathbf{u}_j$ the j -th column of $\mathcal{M}_2$. Let

$$\mathcal{A} := \left\{ i \in [p] : \text{There exists } k \in [n] \text{ such that } \mathbf{y}_{kp+i} = \sum_{\ell=p+1}^{kp+i-1} \gamma_{\ell} \mathbf{y}_{\ell} \text{ for some } \gamma_{\ell} \in \mathbb{R} \text{ and} \right.$$

$$\left. \mathbf{y}_{(k-1)p+i} \neq \sum_{\ell=1}^{(k-1)p+i-1} \delta_{\ell} \mathbf{y}_{\ell} \text{ for all } \delta_{\ell} \in \mathbb{R} \right\}$$

and

$$\mathcal{B} := \left\{ i \in [p] : \mathbf{u}_{(n+1)p+i} = \sum_{\ell=p+1}^{(n+1)p+i-1} \gamma_{\ell} \mathbf{u}_{\ell} \text{ for some } \gamma_{\ell} \in \mathbb{R} \text{ and} \right.$$

$$\left. \mathbf{y}_{np+i} \neq \sum_{\ell=1}^{np+i-1} \delta_{\ell} \mathbf{y}_{\ell} \text{ for all } \delta_{\ell} \in \mathbb{R} \right\}.$$

(See Remark 1.2 for an interpretation of the sets $\mathcal{A}$ and $\mathcal{B}$.) Then the following statements are equivalent:

- (1) There exists a minimal representing matrix measure μ for L such that $\text{mult}_{\mu} t = m$.
- (2) $\text{card } \mathcal{A} \leq m \leq \text{card}(\mathcal{A} \cup \mathcal{B})$.

Moreover, writing $\mathcal{B} := \{i_1, i_2, \dots, i_{\text{card } \mathcal{B}}\}$ and $\{j_1, j_2, \dots, j_{p-\text{card } \mathcal{B}}\} := [p] \setminus \mathcal{B}$, the representing measure is unique if and only if

$$(1.4) \quad \begin{aligned} \mathcal{B} &= \left\{ i \in [p] : \mathbf{y}_{np+i} \neq \sum_{\ell=1}^{np+i-1} \delta_\ell \mathbf{y}_\ell \text{ for all } \delta_\ell \in \mathbb{R} \right\}, \\ m &= \text{card}(\mathcal{A} \cup \mathcal{B}), \\ \text{Ker}(T_{i+j-2})_{\substack{i=1,\dots,n, \\ j=2,\dots,n+1}} &\subseteq \text{Ker}(\mathbf{u}_{(n+1)p+i_1} \mathbf{u}_{(n+1)p+i_2} \cdots \mathbf{u}_{(n+1)p+i_{\text{card } \mathcal{B}}})^T, \\ \text{Ker}(T_{i+j-2})_{\substack{i=1,\dots,n, \\ j=2,\dots,n+1}} &\subseteq \text{Ker}(\mathbf{u}_{(n+1)p+j_1} \mathbf{u}_{(n+1)p+j_2} \cdots \mathbf{u}_{(n+1)p+j_{p-\text{card } \mathcal{B}}})^T. \end{aligned}$$

(See Remark 1.3 for an interpretation of the uniqueness conditions.)

Remark 1.2 (Interpretation of the sets $\mathcal{A}$ and $\mathcal{B}$). The sets $\mathcal{A}$ and $\mathcal{B}$ describe how column relations determine the possible multiplicity of the prescribed point t .

The columns are indexed by the vector monomials $(x-t)^k e_i$, where $e_1, \dots, e_p$ are the standard basis vectors of $\mathbb{R}^p$. The pairs (k, i) are ordered first by increasing k and, for each fixed k , by increasing i . Thus, omitting the first block column from a dependence relation means that the corresponding vector polynomial is divisible by $x-t$.

The set $\mathcal{A}$ records new relations of this form that already occur in $\mathcal{M}_1$. The independence condition at the preceding block column ensures that, after division by $x-t$, the resulting polynomial is not itself a column relation. This polynomial therefore has positive squared norm with respect to any representing measure, whereas its product with $x-t$ has zero squared norm. Its nonzero contribution must consequently come from the mass at t . These relations account for the unavoidable contribution of $\text{card } \mathcal{A}$ to the multiplicity of t .

The set $\mathcal{B}$ concerns indices whose columns remain independent through the last block column of $\mathcal{M}_1$ and records the possible increase in multiplicity beyond the contribution $\text{card } \mathcal{A}$ already forced by the given moments. More precisely, Theorem 1.1 states that, among minimal representing measures, the multiplicity of t can take every integer value between $\text{card } \mathcal{A}$ and $\text{card}(\mathcal{A} \cup \mathcal{B})$.

Remark 1.3 (Interpretation of the uniqueness conditions). Uniqueness in Theorem 1.1 is understood among minimal representing measures μ satisfying $\text{mult}_\mu t = m$. The conditions in (1.4) describe when the construction of such a measure leaves no freedom in the resulting moment extension.

The first condition requires $\mathcal{B}$ to coincide with the set of all indices whose column in the last block column of $\mathcal{M}_1$ is independent of its preceding columns. The second requires the prescribed multiplicity to attain its largest possible value, namely $m = \text{card}(\mathcal{A} \cup \mathcal{B})$. Together, these conditions eliminate the freely selectable symmetric block in the extension construction used in the proof.

The two kernel inclusions ensure that any ambiguity in the coefficients used to express the column dependencies does not affect the resulting extension. Although these coefficient representations may be nonunique, they must yield the same completed moments. The resulting moment extension is therefore uniquely determined, as is its associated minimal representing measure.

Remark 1.4. A special case of Theorem 1.1, corresponding to the situation where $\mathcal{M}_1$ is positive definite, was established in [ZZ26, Theorem 1.1]. Under this assumption we have $\mathcal{A} = \emptyset$, and

$$(1.5) \quad \begin{aligned} \text{card } \mathcal{B} &= \text{rank}(T_{i+j-2})_{\substack{i=1,\dots,n, \\ j=2,\dots,n+1}} - (n-1)p \\ &= \text{rank}(S_{i+j-1} - tS_{i+j-2})_{i,j=1}^n - (n-1)p, \end{aligned}$$

where the second equality follows by Proposition 2.8(2), i.e., in the notation of Proposition 2.8 we have

$$(J_\phi \otimes I_p)^{-T} (T_{i+j-2})_{\substack{i=1,\dots,n \\ j=2,\dots,n+1}} (J_\phi \otimes I_p)^{-1} = (S_{i+j-1} - tS_{i+j-2})_{i,j=1}^n,$$

where $J_\phi : \mathbb{R}^n \rightarrow \mathbb{R}^n$. Hence,

$$\text{rank}(T_{i+j-2})_{\substack{i=1,\dots,n \\ j=2,\dots,n+1}} = \text{rank}(S_{i+j-1} - tS_{i+j-2})_{i,j=1}^n.$$

Using (1.5), Theorem 1.1(2) coincides with the statement of [ZZ26, Theorem 1.1(2)] in the case $\mathcal{M}_1$ is positive definite. We also note that, in order to state [ZZ26, Theorem 1.1], it was sufficient to use the localizing matrix $(S_{i+j-1} - tS_{i+j-2})_{i,j=1}^n$, since $\mathcal{M}_1$ was assumed to be invertible. In the singular case, however, it is more convenient to work with the matrices $\mathcal{M}_1$ and $\mathcal{M}_2$, as this makes it possible to capture column dependencies within $\mathcal{M}_1$ as well. These dependencies are essential for the formulation of Theorem 1.1.

In the case $m = 0$, we can add another equivalence to Theorem 1.1.

Theorem 1.5. *Let $n, p \in \mathbb{N}$ and $L : \mathbb{R}[x]_{\leq 2n} \rightarrow \mathbb{S}_p(\mathbb{R})$ be a linear operator, which admits a representing measure. Fix $t \in \mathbb{R}$. Let $T_i := L((x - t)^i)$ for $i \in [0; 2n]$,*

$$(1.6) \quad \mathcal{M}_1 := (T_{i+j-2})_{i,j=1}^{n+1}, \quad \mathcal{M}_2 := (T_{i+j-2})_{i,j=1}^n \quad \text{and} \quad \mathcal{M}_3 := (T_{i+j})_{i,j=1}^n.$$

Denote by $\mathbf{y}_j$ the j -th column of $\mathcal{M}_1$. Let

$$\mathcal{A} := \left\{ i \in [p] : \text{There exists } k \in [n] \text{ such that } \mathbf{y}_{kp+i} = \sum_{\ell=p+1}^{kp+i-1} \gamma_\ell \mathbf{y}_\ell \text{ for some } \gamma_\ell \in \mathbb{R} \text{ and} \right. \\ \left. \mathbf{y}_{(k-1)p+i} \neq \sum_{\ell=1}^{(k-1)p+i-1} \delta_\ell \mathbf{y}_\ell \text{ for all } \delta_\ell \in \mathbb{R} \right\}.$$

The following statements are equivalent:

- (1) There exists a minimal representing matrix measure μ for L such that $\text{mult}_\mu t = 0$.
- (2) $\mathcal{A} = \emptyset$.
- (3) $\text{Ker } \mathcal{M}_2 = \text{Ker } \mathcal{M}_3$.

As a consequence of Theorem 1.5 we obtain the following corollary, which recovers [Sim06, Theorem 3.3] and provides a solution to a strong truncated matrix Hamburger moment problem.

Corollary 1.6. *Let $n_1, n_2, p \in \mathbb{N}$ and $S_i \in \mathbb{S}_p(\mathbb{R})$ for $i \in [-2n_1; 2n_2]$. Then the following statements are equivalent:*

- (1) There exists a measure μ such that $S_i = \int_{\mathbb{R}} x^i d\mu$ for $i \in [-2n_1; 2n_2]$.
- (2) There exists a finitely atomic measure μ such that $S_i = \int_{\mathbb{R}} x^i d\mu$ for $i \in [-2n_1; 2n_2]$.
- (3) There exists a minimal representing measure μ such that

$$S_i = \int_{\mathbb{R}} x^i d\mu \quad \text{for } i \in [-2n_1; 2n_2].$$

- (4) $(S_{i+j})_{i,j=-n_1}^{n_2}$ is positive semidefinite and $\text{Ker}(S_{i+j})_{i,j=-n_1}^{n_2-1} = \text{Ker}(S_{i+j})_{i,j=-n_1+1}^{n_2}$.
- (5) $(S_{i+j})_{i,j=-n_1}^{n_2}$ is positive semidefinite and, for every sequence $\{v_k\}_{k=0}^{n_1+n_2-1} \subset \mathbb{R}^p$,

$$(\text{col}(v_{i+n_1})_{i \in [-n_1; n_2-1]})^T (S_{i+j})_{i,j=-n_1}^{n_2-1} \text{col}(v_{i+n_1})_{i \in [-n_1; n_2-1]} = 0$$

if and only if

$$(\text{col}(v_{i+n_1})_{i \in [-n_1; n_2-1]})^T (S_{i+j+2})_{i,j=-n_1}^{n_2-1} \text{col}(v_{i+n_1})_{i \in [-n_1; n_2-1]} = 0.$$

For an alternative solution of the scalar strong truncated Hamburger moment problem, based on the truncated Hamburger moment problem and the properties of Hankel matrices, see [Zal22b, Theorem 3.1].

In [BKRSV20, Theorem 1.4], the authors solved the scalar version of the Problem in the nonsingular case (i.e., $p = 1$ in (1.1) and the moment matrix $M(n)$ as in (2.2) is positive definite) in terms of symmetric determinantal representations involving moment matrices using convex analysis and algebraic geometry. For an alternative proof using moment theory, and an extension to finitely many prescribed atoms, see [NZ+]. In [ZZ26, Theorem 1.1], we extended [BKRSV20, Theorem 1.4] to the general matrix case (i.e., $p \in \mathbb{N}$). We remark that, in the scalar setting, the restriction to positive definite moment matrices $M(n)$ is natural. Specifically, when $M(n)$ is positive semidefinite but not positive definite, the minimal representing measure is uniquely determined [CF91, Theorems 3.9 and 3.10]. This uniqueness property does not extend to the matrix-valued case, whence a formulation of the problem with positive semidefinite $M(n)$ remains relevant. Furthermore, this variant of the problem is technically more intricate and is addressed in the present paper.

In [FKM24], the authors described, for each $t \in \mathbb{R}$, the set of all possible masses at t among all representing measures for L , addressing a question related to the Problem above. The multivariate analogue of this problem was investigated in [MS24a, MS24b]. The main distinction of our work lies in the fact that we focus on *minimal* representing measures with a *fixed rank* of the mass at t .

The works [DD02, DLR96, DS03] focus on the computation of atoms and masses in matricial Gaussian quadrature rules, where the moment S_{2n+1} is fixed, and consequently, the minimal representing measure is unique. The corresponding formulas are expressed in terms of the zeros of the associated orthogonal matrix polynomial. A distinctive feature of our results is that we do not fix S_{2n+1} *a priori*. Instead, we characterize the existence of a suitable S_{2n+1} that yields a minimal representing measure containing a prescribed atom with a prescribed multiplicity. In the proof, we explicitly construct such an S_{2n+1} with the required properties, ensuring that the extended moment matrix $M(n+1)$ satisfies $\text{rank } M(n+1) = \text{rank } M(n)$ and admits an appropriate block column relation (see Subsection 2.4).

The techniques employed in [Sim06] to establish Corollary 1.6 rely on advanced operator-theoretic methods, involving the analysis of self-adjoint extensions of a certain linear operator that is not necessarily defined on the entire finite-dimensional Hilbert space of vector-valued Laurent polynomials. In contrast, the present paper provides a constructive, linear-algebraic proof, in the sense that the representing measures can be obtained explicitly by following the steps outlined in the proof of Theorem 1.1.

1.1. Reader's guide. In Section 2 we introduce the notation and some preliminary results. In Section 3 we prove the implication (2) $\Rightarrow$ (1) of Theorem 1.1. In the last paragraph of the section we also prove the moreover part of Theorem 1.1. In Section 4 we prove the remaining implication (1) $\Rightarrow$ (2). Section 5 is devoted to the proof of Theorem 1.5, while Section 6 to Corollary 1.6. Finally, in Section 7 we demonstrate the statement of Theorem 1.1 on a numerical example.

2. PRELIMINARIES

Let $m, m_1, m_2 \in \mathbb{N}$. We write $M_{m_1 \times m_2}(\mathbb{R})$ for the set of $m_1 \times m_2$ real matrices and $M_m(\mathbb{R}) \equiv M_{m \times m}(\mathbb{R})$ for short. For a matrix $A \in M_{m_1 \times m_2}(\mathbb{R})$ we call the linear span of its columns a **column space** and denote it by $\mathcal{C}(A)$. We denote by I_m the identity $m \times m$ matrix and by $\mathbf{0}_{m_1 \times m_2}$ the zero $m_1 \times m_2$ matrix, while $\mathbf{0}_m \equiv \mathbf{0}_{m \times m}$ for short. We use $M_m(\mathbb{R}[x])$ to denote $m \times m$ matrices over $\mathbb{R}[x]$. The elements of $M_m(\mathbb{R}[x])$ are called **matrix polynomials**.

Let $p \in \mathbb{N}$. For $A \in \mathbb{S}_p(\mathbb{R})$ the notation $A \succeq 0$ (resp. $A \succ 0$) means A is positive semidefinite (psd) (resp. positive definite (pd)). We use $\mathbb{S}_p^{\succeq 0}(\mathbb{R})$ for the subset of all psd matrices in $\mathbb{S}_p(\mathbb{R})$.

Given a polynomial $p(x) \in \mathbb{R}[x]$, we write $\mathcal{Z}(p(x)) := \{x \in \mathbb{R} : p(x) = 0\}$ for the set of its zeros and by $\mathcal{Z}_{\mathbb{R}}(p(x)) := \mathcal{Z}(p(x)) \cap \mathbb{R}$ the set of its real zeros. For $a \in \mathcal{Z}_{\mathbb{R}}(p(x))$ we denote by $\text{mult}_{p(x)} a$ the vanishing order of a as a zero of $p(x)$.

Let $A_k, k \in [i; j]$ be given matrices. We use

$$\begin{aligned} \text{row}(A_k)_{k \in [i; j]} &\equiv \text{row}(A_i, A_{i+1}, \dots, A_j), \\ \text{col}(A_k)_{k \in [i; j]} &\equiv \text{col}(A_i, A_{i+1}, \dots, A_j) \end{aligned}$$

for the row vector and the column vector with entries A_k , respectively. If $[i; j] = \emptyset$, the expressions $\text{row}(A_k)_{k \in [i; j]}$ and $\text{col}(A_k)_{k \in [i; j]}$ are understood as empty matrices.

2.1. Matrix measures. Let $\text{Bor}(\mathbb{R})$ be the Borel σ -algebra of $\mathbb{R}$. We call

$$\mu = (\mu_{ij})_{i,j=1}^p : \text{Bor}(\mathbb{R}) \rightarrow \mathbb{S}_p(\mathbb{R})$$

a $p \times p$ **Borel matrix-valued measure** supported on $\mathbb{R}$ (or **positive $\mathbb{S}_p(\mathbb{R})$ -valued measure**) if

- (1) $\mu_{ij} : \text{Bor}(\mathbb{R}) \rightarrow \mathbb{R}$ is a real measure for every $i, j \in [p]$ and
- (2) $\mu(\Delta) \succeq 0$ for every $\Delta \in \text{Bor}(\mathbb{R})$.

A positive $\mathbb{S}_p(\mathbb{R})$ -valued measure μ is **finitely atomic** if there exists a finite set $M \in \text{Bor}(\mathbb{R})$ such that $\mu(\mathbb{R} \setminus M) = \mathbf{0}_p$ or equivalently, $\mu = \sum_{j=1}^{\ell} \delta_{x_j} A_j$ for some $\ell \in \mathbb{N}$, $x_j \in \mathbb{R}$, $A_j \in \mathbb{S}_p^{\succeq 0}(\mathbb{R})$. We say μ is **r -atomic** if $\sum_{j=1}^{\ell} \text{rank } A_j = r$.

Let μ be a positive $\mathbb{S}_p(\mathbb{R})$ -valued measure and $\tau := \text{tr}(\mu) = \sum_{i=1}^p \mu_{ii}$ denote its trace measure. A polynomial $f \in \mathbb{R}[x]$ is μ -integrable if $f \in L^1(\tau)$. We define its integral by

$$\int_{\mathbb{R}} f d\mu := \left(\int_{\mathbb{R}} f d\mu_{ij} \right)_{i,j=1}^p.$$

2.2. Riesz mapping. One can define L as in (1.1) by a sequence of its values on monomials $x^i, i \in [0; 2n]$. Throughout the paper we will denote these values by $S_i := L(x^i)$. If $\mathcal{S} := (S_0, S_1, \dots, S_{2n}) \in (\mathbb{S}_p(\mathbb{R}))^{2n+1}$ is given, then we denote the corresponding linear mapping on $\mathbb{R}[x]_{\leq 2n}$ by $L_{\mathcal{S}}$ and call it a **Riesz mapping of $\mathcal{S}$** .

2.3. Moment matrix and its column dependencies. For $n \in \mathbb{N}$ and

$$(2.1) \quad \mathcal{S} \equiv \mathcal{S}^{(2n)} := (S_0, S_1, \dots, S_{2n}) \in (\mathbb{S}_p(\mathbb{R}))^{2n+1},$$

we denote by

$$(2.2) \quad M(n) \equiv M_{\mathcal{S}}(n) := (S_{i+j-2})_{i,j=1}^{n+1} = \begin{matrix} & \begin{matrix} 1 & X & X^2 & \cdots & X^n \end{matrix} \\ \begin{matrix} 1 \\ X \\ X^2 \\ \vdots \\ X^n \end{matrix} & \begin{pmatrix} S_0 & S_1 & S_2 & \cdots & S_n \\ S_1 & S_2 & \ddots & \ddots & S_{n+1} \\ S_2 & \ddots & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & S_{2n-1} \\ S_n & S_{n+1} & \cdots & S_{2n-1} & S_{2n} \end{pmatrix} \end{matrix}$$

the corresponding n -th **truncated moment matrix**. We write

$$(2.3) \quad X^i := \text{col}(S_{i+\ell})_{\ell \in [0; n]}.$$

Remark 2.1. Further, we will also use $X_{\mathcal{S}}^i$ to denote the i -th block column of the largest moment matrix, fully determined by $\mathcal{S}$.

In the notation (2.3), we have $M(n) = \text{row}(X^i)_{i \in [0;n]}$. Let us denote the columns of each block X^i by $x_j^{(i)}$, i.e.,

$$(2.4) \quad X^i = \begin{pmatrix} x_1^{(i)} & x_2^{(i)} & \cdots & x_p^{(i)} \end{pmatrix} = \text{row}(x_j^{(i)})_{j \in [p]}.$$

For $i \in [0; n]$ let $\text{Dep}(X^i) \equiv \text{Dep}(X_S^i)$ be the set of all indices $j \in [p]$ for which the column $x_j^{(i)}$ is in the span of the previous columns of the matrix $M(n)$, i.e.,

$$(2.5) \quad \text{Dep}(X^i) := \left\{ j \in [p] : x_j^{(i)} \in \mathcal{C} \left(\text{row}(X^k)_{k \in [0;i-1]} \quad \text{row}(x_k^{(i)})_{k \in [j-1]} \right) \right\}.$$

Similarly, for $i \in [n]$ let $\text{Dep}_1(X^i) \equiv \text{Dep}_1(X_S^i)$ be the set of all indices $j \in [p]$ for which the column $x_j^{(i)}$ is in the span of all previous columns of the matrix $M(n)$ *starting from the block column* X^1 , i.e.,

$$(2.6) \quad \text{Dep}_1(X^i) := \left\{ j \in [p] : x_j^{(i)} \in \mathcal{C} \left(\text{row}(X^k)_{k \in [i-1]} \quad \text{row}(x_k^{(i)})_{k \in [j-1]} \right) \right\}.$$

Define $\text{Dep}_1(X^0) := \emptyset$. Further, let

$$(2.7) \quad \begin{aligned} \text{DepN}(X^0) &:= \text{Dep}(X^0), \\ \text{DepN}(X^i) &:= \text{Dep}(X^i) \setminus \text{Dep}(X^{i-1}) \quad \text{for } i \in [n], \\ \text{DepN}_1(X^0) &:= \emptyset, \\ \text{DepN}_1(X^i) &:= \text{Dep}_1(X^i) \setminus \text{Dep}(X^{i-1}) \quad \text{for } i \in [n], \\ \text{DepN}_0(X^i) &:= \text{DepN}(X^i) \setminus \text{DepN}_1(X^i) \quad \text{for } i \in [0; n]. \end{aligned}$$

Here, the letter N stands for “new”. For $i \in [n]$, $\text{DepN}(X^i)$ records the indices at which a new column dependence appears relative to the preceding block column X^{i-1} . Among these indices, $\text{DepN}_1(X^i)$ consists of those for which the corresponding column lies in the span of its preceding columns even when the block column X^0 is omitted, whereas $\text{DepN}_0(X^i)$ consists of those for which every such representation requires at least one column from X^0 .

The following lemma captures relations between the above sets.

Lemma 2.2. *For every $i \in [0; n]$, we have*

$$\begin{aligned} \text{Dep}_1(X^i) &\subseteq \text{Dep}(X^i), \\ \text{DepN}_\ell(X^i) &\subseteq \text{DepN}(X^i) \subseteq \text{Dep}(X^i), \quad \ell = 0, 1, \\ \text{DepN}_1(X^i) &= \text{Dep}_1(X^i) \cap \text{DepN}(X^i), \\ \text{DepN}_0(X^i) &= \text{DepN}(X^i) \setminus \text{Dep}_1(X^i), \end{aligned}$$

and consequently

$$\text{DepN}(X^i) = \text{DepN}_0(X^i) \sqcup \text{DepN}_1(X^i).$$

Proof. The assertions follow directly from (2.5), (2.6) and (2.7). $\square$

For a concrete illustration of these notions, see the example in Section 7, where the corresponding sets $\text{Dep}(X_{\mathcal{T}}^i)$, $\text{Dep}_1(X_{\mathcal{T}}^i)$, $\text{DepN}(X_{\mathcal{T}}^i)$, $\text{DepN}_0(X_{\mathcal{T}}^i)$ and $\text{DepN}_1(X_{\mathcal{T}}^i)$ are computed explicitly for the sequence $\mathcal{T}$ considered there.

2.4. Evaluation of a matrix polynomial on $M(n)$. Let $\mathcal{S}$ be a sequence as in (2.1) and $M(n)$ its corresponding moment matrix. Given a matrix polynomial

$$(2.8) \quad P(x) = \sum_{i=0}^n x^i P_i \in M_p(\mathbb{R}[x]),$$

the **evaluation** $P(X)$ on $M(n)$ is a matrix, obtained by replacing each monomial of P by the corresponding column of $M(n)$ and multiplying it by the matrix coefficients P_i from the right, i.e.,

$$P(X) \equiv P(X_{\mathcal{S}}) := \sum_{i=0}^n X_{\mathcal{S}}^i P_i = X_{\mathcal{S}}^0 P_0 + X_{\mathcal{S}}^1 P_1 + \cdots + X_{\mathcal{S}}^n P_n \in M_{(n+1)p \times p}(\mathbb{R}),$$

where X^i are as in (2.3) above. If $P(X) = \mathbf{0}_{(n+1)p \times p}$, then we say $P(x)$ is a **block column relation** of $M(n)$.

Given a matrix polynomial $P(x)$ as in (2.8), we write

$$(2.9) \quad \text{coef}(P(x)) := \text{col}(P_i)_{i \in [0;n]}$$

to denote the vector of its matrix coefficients.

A moment matrix $M(n)$ is **block–recursively generated** if, for every block column relation $P(x) = \sum_{i=0}^j x^i P_i \in M_p(\mathbb{R}[x])$, $0 \leq j < n$, of $M(n)$, the matrix polynomial $(xP)(x) = \sum_{i=0}^j x^{i+1} P_i$ is also a block column relation of $M(n)$.

2.5. Solution to the truncated matricial Hamburger moment problem.

Theorem 2.3 ([BW11, Theorem 2.7.6]). *Let $n, p \in \mathbb{N}$ and let $\mathcal{S} := (S_0, S_1, \dots, S_{2n}) \in (\mathbb{S}_p(\mathbb{R}))^{2n+1}$ be a sequence with a moment matrix $M(n)$. Then the following statements are equivalent:*

- (1) *There exists a representing matrix measure for $\mathcal{S}$.*
- (2) *There exists a $(\text{rank } M(n))$ –atomic representing matrix measure for $\mathcal{S}$.*
- (3) *$M(n)$ is positive semidefinite and $\mathcal{C}(\text{col}(S_{n+i})_{i \in [n]}) \subseteq \mathcal{C}(M(n-1))$.*

Remark 2.4. The truncated matrix Hamburger moment problem was also considered in [Bol96, Dym89, DFKMT09].

The following lemma connects the support $\text{supp}(\mu)$ of a representing measure μ for $\mathcal{S}$ as in Theorem 2.3 with block column relations of $M(n)$.

Lemma 2.5. *Let $n, p \in \mathbb{N}$ and let $\mathcal{S} := (S_0, S_1, \dots, S_{2n}) \in (\mathbb{S}_p(\mathbb{R}))^{2n+1}$ be a sequence with a moment matrix $M(n)$ and a representing measure μ . Assume that $H(x) = \sum_{i=0}^n x^i H_i \in M_p(\mathbb{R}[x])$ is a block column relation of $M(n)$ such that H_n is invertible. Then*

$$(2.10) \quad \text{supp}(\mu) \subseteq \mathcal{Z}(\det H(x)) \quad \text{and} \quad \text{mult}_{\mu} \xi \leq \text{mult}_{\det H(x)} \xi \quad \text{for each } \xi \in \mathbb{R}.$$

Proof. The inclusion $\text{supp}(\mu) \subseteq \mathcal{Z}(\det H(x))$ is [KT22, Lemma 5.53]. It remains to prove the second assertion in (2.10). Let $L \equiv L_{\mathcal{S}} : \mathbb{R}[x]_{\leq 2n} \rightarrow \mathbb{S}_p(\mathbb{R})$ be the Riesz mapping of $\mathcal{S}$. We define a functional

$$\Lambda_L : \mathbb{S}_p(\mathbb{R}[x]) \rightarrow \mathbb{R}, \quad \Lambda_L((F_{jk})_{j,k}) := \sum_{1 \leq j \leq p} L(F_{jj})_{j,j} + \sum_{1 \leq j < k \leq p} 2L(F_{jk})_{j,k}$$

By the real version of [MS24a, Proposition 4.5], representing measures for L coincide with representing measures for Λ_L . Here, $\tilde{\mu}$ is a representing measure for Λ_L if and only if $\Lambda_L(F) = \int_{\mathbb{R}} \text{tr}(F(x)\Phi(x))d\tilde{\tau}$, where $\tilde{\tau} = \text{tr}(\tilde{\mu})$ is a trace measure and $\Phi(x)$ is so-called Radon-Nikodym

matrix of $\tilde{\mu}$ (see [Sch87, Section 1]). Let $\text{coef}(H(x))$ be as in (2.9). Since $H(x)$ is a block column relation of $M(n)$, in particular we have that

$$\Lambda_L(H(x)(H(x))^T) = \text{tr}(\text{coef}(H(x))^T M(n) \text{coef}(H(x))) = 0,$$

where the first equality follows by a short computation as in [MS24b, Lemma 3.2]. By the real version of [MS24a, Lemma 7.7] (using $P(x)$ being constantly equal to the identity function on $\mathbb{R}$), in particular it follows that

$$\text{rank } \mu(\{\xi\}) \leq \dim \text{Ker}(H(\xi)H(\xi)^T) = \dim \text{Ker } H(\xi)$$

for every $\xi \in \text{supp}(\mu)$. To conclude the proof we need to show that

$$(2.11) \quad \dim \text{Ker } H(\xi) \leq \text{mult}_{\det H(x)} \xi.$$

By [GLR82, Theorem 1.1], $I_{np}x - C = E(x)(H_n^{-1}H(x) \oplus I_{(n-1)p})F(x)$, where $C \in M_{np}(\mathbb{R})$ is the companion matrix of $H_n^{-1}H(x)$ and $E(x), F(x) \in M_{np}(\mathbb{R}[x])$ are matrix polynomials with constant nonzero determinant. Hence, $\dim \text{Ker}(I_{np}\xi - C) = \dim \text{Ker } H(\xi)$. Since the algebraic multiplicity of ξ as the eigenvalue of the matrix C is not smaller than the dimension of the eigenspace of C corresponding to ξ , (2.11) follows. $\square$

Proposition 2.6. *Let $n, p \in \mathbb{N}$ and let $\mathcal{S} := (S_0, S_1, \dots, S_{2n}) \in (\mathbb{S}_p(\mathbb{R}))^{2n+1}$ be a sequence with a representing measure μ . Then $M(n)$ is block-recursively generated.*

Proof. By the matricial Richter-Tchakaloff's theorem [MS24a, Theorem 5.1], there exists a finitely atomic representing measure $\tilde{\mu}$ for $\mathcal{S}$. Let $\mathcal{S}^{(\infty)} = (S_i)_{i \in \mathbb{N}_0}$ be an infinite sequence, obtained by $S_i := \int_{\mathbb{R}} x^i d\tilde{\mu}$. Let $M(\infty) = (S_{i+j-2})_{i,j \in \mathbb{N}}$ be the corresponding Hankel matrix. Let $P(x) = \sum_{i=0}^j x^i P_i \in M_p(\mathbb{R}[x])$, $0 \leq j < n$, be a block column relation of $M(n)$. By [KT22, Lemma 5.21], P is a block column relation of $M(\infty)$, where the definition of the evaluation $P(X)$ from Subsection 2.4 extends from $M(n)$ to $M(\infty)$ in a natural way. By [KT22, Lemma 5.15], $(P \cdot xI_p) = xI_p \cdot P = xP$ is a column relation of $M(\infty)$ and in particular, of $M(n)$. $\square$

Corollary 2.7. *Let $n, p \in \mathbb{N}$ and let $\mathcal{S} := (S_0, S_1, \dots, S_{2n}) \in (\mathbb{S}_p(\mathbb{R}))^{2n+1}$ be a sequence with a representing measure μ . For $i \in [0; n]$, let $\text{Dep}(X^i)$, $\text{Dep}_1(X^i)$ and $\text{DepN}(X^i)$ be as in (2.5), (2.6) and (2.7), respectively. Then*

$$\text{Dep}(X^i) \subseteq \text{Dep}_1(X^{i+1}) \subseteq \text{Dep}(X^{i+1}) \quad \text{for } i \in [0; n-1].$$

Consequently,

$$0 \leq \text{card } \text{Dep}(X^i) \leq \text{card } \text{Dep}(X^{i+1}) \leq p \quad \text{for } i \in [0; n-1],$$

and

$$\text{Dep}(X^i) = \bigsqcup_{k=0}^i \text{DepN}(X^k) \quad \text{for } i \in [0; n].$$

Proof. By Proposition 2.6, $M(n)$ is block-recursively generated. Fix $i \in [0; n-1]$ and $j \in \text{Dep}(X^i)$. Then

$$x_j^{(i)} = \sum_{k=0}^{i-1} X^k a_k + \sum_{\ell=1}^{j-1} \alpha_\ell x_\ell^{(i)}$$

for some $a_k \in \mathbb{R}^p$ and $\alpha_\ell \in \mathbb{R}$. Equivalently, letting $e_1, \dots, e_p$ denote the standard basis vectors of $\mathbb{R}^p$, the matrix polynomial

$$P(x) := x^i e_j e_1^T - \sum_{k=0}^{i-1} x^k a_k e_1^T - \sum_{\ell=1}^{j-1} \alpha_\ell x^\ell e_\ell e_1^T$$

is a block column relation of $M(n)$. Since $i < n$ and $M(n)$ is block-recursively generated, $xP(x)$ is also a block column relation of $M(n)$. Taking the first column of the matrix relation $(xP)(X) = \mathbf{0}_{(n+1)p \times p}$ and using the definition of $P(x)$ gives

$$x_j^{(i+1)} = \sum_{k=0}^{i-1} X^{k+1} a_k + \sum_{\ell=1}^{j-1} \alpha_\ell x_\ell^{(i+1)}.$$

This expresses $x_j^{(i+1)}$ in terms of its preceding columns without using X^0 , so $j \in \text{Dep}_1(X^{i+1})$. Hence $\text{Dep}(X^i) \subseteq \text{Dep}_1(X^{i+1})$. The inclusion $\text{Dep}_1(X^{i+1}) \subseteq \text{Dep}(X^{i+1})$ follows from Lemma 2.2.

The cardinality inequalities follow immediately. Finally, the disjoint union formula follows from the nesting of the sets $\text{Dep}(X^i)$ and the definition of $\text{DepN}(X^i)$. $\square$

2.6. Change of basis on block columns and rows of the moment matrix. Let $\mathcal{S} \equiv \mathcal{S}^{(2n)}$ be as in (2.1) and let $t \in \mathbb{R}$. Let $\phi(x) = x - t$ be an invertible affine linear transformation and let

$$\mathcal{T} \equiv \mathcal{T}^{(2n)} \equiv (T_0, T_1, \dots, T_{2n})$$

be a sequence, defined for each $i \in [0; 2n]$ by

$$T_i := L_{\mathcal{S}}((x - t)^i) = L_{\mathcal{S}}\left(\sum_{\ell=0}^i \binom{i}{\ell} (-1)^\ell x^{i-\ell} t^\ell\right) = \sum_{\ell=0}^i \binom{i}{\ell} (-1)^\ell S_{i-\ell} t^\ell.$$

The following proposition is a version of [CF05, Proposition 1.9] for matricial sequences.

Proposition 2.8. *Suppose $\phi, \mathcal{S}, \mathcal{T}$ are as above and let $M_{\mathcal{S}}(n), M_{\mathcal{T}}(n)$ be the corresponding moment matrices. We denote by $X_{\mathcal{S}}^i$ and $X_{\mathcal{T}}^i$ the i -th block column of $M_{\mathcal{S}}(n)$ and $M_{\mathcal{T}}(n)$, respectively. Let $J_\phi : \mathbb{R}^{n+1} \rightarrow \mathbb{R}^{n+1}$ be a linear map, given by*

$$J_\phi \text{coef}(p(x)) := \text{coef}(p(x - t)),$$

where coef is as in (2.9). We identify J_ϕ with its matrix with respect to the standard basis of $\mathbb{R}^{n+1}$. Thus,

$$J_\phi = (a_{ij})_{i,j=1}^{n+1}, \quad a_{ij} = \begin{cases} \binom{j-1}{i-1} (-t)^{j-i}, & i \leq j, \\ 0, & i > j. \end{cases}$$

We denote by $J_\phi \otimes I_p \in M_{(n+1)p}(\mathbb{R})$ the Kronecker product of J_ϕ and I_p . In particular, for $P(x) = \sum_{j=0}^n x^j P_j \in M_p(\mathbb{R}[x])$,

$$(J_\phi \otimes I_p) \text{coef}(P(x)) = \text{col} \left(\sum_{j=i}^n \binom{j}{i} (-t)^{j-i} P_j \right)_{i \in [0; n]}.$$

Then the following statements are true:

- (1) $(J_\phi \otimes I_p) \text{coef}(P(x)) = \text{coef}((P \circ \phi)(x)) = \text{coef}(P(x - t))$.
- (2) $M_{\mathcal{T}}(n) = (J_\phi \otimes I_p)^T M_{\mathcal{S}}(n) (J_\phi \otimes I_p)$.
- (3) J_ϕ is invertible and, in particular, $J_\phi \otimes I_p$ is invertible.
- (4) $M_{\mathcal{T}}(n) \succeq 0 \Leftrightarrow M_{\mathcal{S}}(n) \succeq 0$.
- (5) $\text{rank } M_{\mathcal{T}}(n) = \text{rank } M_{\mathcal{S}}(n)$.
- (6) For $P(x) = \sum_{i=0}^n x^i P_i \in M_p(\mathbb{R}[x])$, we have $P(X_{\mathcal{T}}) = (J_\phi \otimes I_p)^T ((P \circ \phi)(X_{\mathcal{S}}))$.

Proof. Straightforward. $\square$

Corollary 2.9. *Suppose $n, p \in \mathbb{N}$, $\mathcal{S}, \mathcal{T}, M_{\mathcal{S}}(n)$ and $M_{\mathcal{T}}(n)$ are as above. We denote by $X_{\mathcal{S}}^i$ and $X_{\mathcal{T}}^i$ the i -th block column of $M_{\mathcal{S}}(n)$ and $M_{\mathcal{T}}(n)$, respectively. Assume that $P_{\mathcal{T}}(x) = \sum_{i=0}^n x^i P_i \in M_p(\mathbb{R}[x])$ is a block column relation of $M_{\mathcal{T}}(n)$. Then $P_{\mathcal{S}}(x) := P_{\mathcal{T}}(x - t)$ is a block column relation of $M_{\mathcal{S}}(n)$.*

Proof. Let ϕ and J_ϕ be as in Proposition 2.8. By (3) and (6) of Proposition 2.8, the evaluation $P_S(X_S)$ is equal to $(J_\phi \otimes I_p)^{-T} P_T(X_T)$, which implies the statement of the corollary. $\square$

Proposition 2.10. *Let $n, p \in \mathbb{N}$, $\mathcal{S}$ and $\mathcal{T}$ are as in (2.1) and (2.6), and let $M_S(n)$ and $M_T(n)$ be the corresponding moment matrices. Then $M_S(n)$ is block-recursively generated if and only if $M_T(n)$ is block-recursively generated.*

Proof. Assume the notation from Proposition 2.8. By symmetry it suffices to prove that if $M_S(n)$ is block-recursively generated, it follows that $M_T(n)$ is block-recursively generated. Assume that $M_S(n)$ is block-recursively generated. Let $P(x) = \sum_{i=0}^j x^i P_i \in M_p(\mathbb{R}[x])$, $0 \leq j < n$, be a block column relation of $M_T(n)$. By Proposition 2.8(6), $(P \circ \phi)(x)$ is a block column relation of $M_S(n)$. Since $M_S(n)$ is block-recursively generated, it follows that $(xP \circ \phi)$ is a block column relation of $M_S(n)$. By Proposition 2.8(6), xP is a block column relation of $M_T(n)$, whence $M_T(n)$ is block-recursively generated. $\square$

2.7. Determinant of a matrix polynomial. The following lemma, which will be essentially used in the proofs of our main results, is a refined version of [ZZ26, Lemma 3.1].

Lemma 2.11. *Let $p, n \in \mathbb{N}$, $t \in \mathbb{R}$ and*

$$H(x) = \sum_{i=0}^n (x-t)^i H_i \in M_p(\mathbb{R}[x]_{\leq n})$$

be a nonzero matrix polynomial with invertible H_n . For $k \in [0; n]$ we define

$$W_k := \bigcap_{i=0}^k \text{Ker } H_i \quad \text{and} \quad s_k := \dim W_k.$$

Then

$$(2.12) \quad \det H(x) = (x-t)^{\sum_{k=0}^{n-1} s_k} \cdot g(x),$$

where $0 \neq g(x) \in \mathbb{R}[x]$.

Proof. Note that

$$W_n \subseteq W_{n-1} \subseteq \dots \subseteq W_0 \subseteq \mathbb{R}^p \quad \text{and} \quad 0 \leq s_n \leq s_{n-1} \leq \dots \leq s_0 \leq p.$$

Clearly, since H_n is invertible, $s_n = 0$. Let $\{b_1, b_2, \dots, b_p\}$ be a basis of $\mathbb{R}^p$ such that for each $i \in [0; n-1]$ the set $\{b_1, b_2, \dots, b_{s_i}\}$ is a basis of W_i . Let us define an invertible matrix

$$B := (b_1 \ b_2 \ \dots \ b_p) \in M_p(\mathbb{R}).$$

For $i \in [0; n]$ define matrices

$$(2.13) \quad \tilde{H}_i := H_i B \equiv \begin{pmatrix} \mathbf{0}_{p \times s_i} & \tilde{h}_i^{(s_i+1)} & \dots & \tilde{h}_i^{(p)} \end{pmatrix},$$

where the first s_i columns of $\tilde{H}_i$ are zero due to $b_1, b_2, \dots, b_{s_i} \in \text{Ker } H_i$. Define a matrix polynomial

$$\tilde{H}(x) := \sum_{i=0}^n (x-t)^i \tilde{H}_i = H(x)B \in M_p(\mathbb{R}[x]).$$

By (2.13), it follows that the first s_{n-1} columns of $\tilde{H}(x)$ are of the form

$$(x-t)^n \begin{pmatrix} \tilde{h}_n^{(1)} & \tilde{h}_n^{(2)} & \dots & \tilde{h}_n^{(s_{n-1})} \end{pmatrix},$$

while for every $k \in [n-1]$ the columns $[s_k+1; s_{k-1}]$ of $\tilde{H}(x)$ are equal to

$$(x-t)^k \sum_{i=0}^{n-k} (x-t)^i \begin{pmatrix} \tilde{h}_{i+k}^{(s_k+1)} & \tilde{h}_{i+k}^{(s_k+2)} & \dots & \tilde{h}_{i+k}^{(s_{k-1})} \end{pmatrix}.$$

Observe that the first s_{n-1} columns of $\tilde{H}(x)$ have a common factor $(x-t)^n$, the next $s_{n-2} - s_{n-1}$ columns a common factor $(x-t)^{n-1}$, etc. Using this observation and upon factoring the determinant of $\tilde{H}(x)$ column-wise, we obtain

$$\begin{aligned} \det H(x) &= \frac{\det \tilde{H}(x)}{\det B} = (x-t)^{ns_{n-1}} (x-t)^{(n-1)(s_{n-2}-s_{n-1})} \dots (x-t)^{s_0-s_1} g(x) \\ &= (x-t)^{ns_{n-1} + \sum_{i=1}^{n-1} i(s_{i-1}-s_i)} g(x) \\ &= (x-t)^{\sum_{k=0}^{n-1} s_k} g(x), \end{aligned}$$

which proves (2.12). Since H_n is invertible, $\det H(x) \neq 0$, and thus $g(x) \neq 0$ also holds. $\square$

2.8. Inclusion of kernels. The following technical lemma will be used in the proof of the implication (1) $\Rightarrow$ (2) of Theorem 1.1.

Lemma 2.12. *Let $p \in \mathbb{N}$ and $A, B \in \mathbb{S}_p(\mathbb{R})$ such that $A \succeq B \succeq 0$. Then $\text{Ker } A \subseteq \text{Ker } B$.*

Proof. Let $v \in \text{Ker } A$. From $0 = v^T A v \geq v^T B v \geq 0$, it follows that $v^T B v = 0$. Since $B \succeq 0$, it follows from [HJ12, Observation 7.1.6] that $Bv = 0$. Hence, $v \in \text{Ker } B$. $\square$

2.9. Characterization of positive semidefiniteness of a 2×2 block matrix. Let

$$M = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in M_{n+m}(\mathbb{R})$$

be a real matrix where $A \in M_n(\mathbb{R})$, $B \in M_{n \times m}(\mathbb{R})$, $C \in M_{m \times n}(\mathbb{R})$ and $D \in M_m(\mathbb{R})$. The **generalized Schur complement** [Zha05] of A (resp. D) in M is defined by

$$M/A = D - CA^\dagger B \quad (\text{resp. } M/D = A - BD^\dagger C),$$

where $A^\dagger$ (resp. $D^\dagger$) stands for the Moore-Penrose inverse of A (resp. D).

The following theorem is a characterization of psd 2×2 block matrices.

Theorem 2.13 ([Alb69]). *Let*

$$M = \begin{pmatrix} A & B \\ B^T & C \end{pmatrix} \in \mathbb{S}_{n+m}(\mathbb{R})$$

be a real symmetric matrix where $A \in \mathbb{S}_n(\mathbb{R})$, $B \in M_{n \times m}(\mathbb{R})$ and $C \in \mathbb{S}_m(\mathbb{R})$. Then:

(1) *The following conditions are equivalent:*

- (a) $M \succeq 0$.
- (b) $C \succeq 0$, $\mathcal{C}(B^T) \subseteq \mathcal{C}(C)$ and $M/C \succeq 0$.
- (c) $A \succeq 0$, $\mathcal{C}(B) \subseteq \mathcal{C}(A)$ and $M/A \succeq 0$.

(2) *If $M \succeq 0$, then*

$$\text{rank } M = \text{rank } A + \text{rank } M/A = \text{rank } C + \text{rank } M/C.$$

2.10. Extension principle. The following lemma shows that, for a positive semidefinite matrix, any kernel vector of a principal submatrix can be extended by zeros to a kernel vector of the full matrix. We use it in the proof of Theorem 1.5 in Section 5.

Lemma 2.14 ([Zal22a, Lemma 2.4]). *Let $A \in \mathbb{S}_n(\mathbb{R})$ be positive semidefinite, $Q \subseteq [n]$ and $A|_Q$ be the restriction of A to the rows and columns from the set Q . If $A|_Q v = 0$ for a nonzero vector v , then $A\hat{v} = 0$ where $\hat{v}$ is a vector with the only nonzero entries in the rows from Q and such that the restriction $\hat{v}|_Q$ to the rows from Q equals to v .*

2.11. Lemma on the rank of the matrix. Another technical lemma that will be used in the proof of the implication (2) $\Rightarrow$ (1) of Theorem 1.1 is the following.

Lemma 2.15. *Let $m_1, m_2, c, k, \ell \in \mathbb{N}$ and $A_1 \in M_{m_1 \times k}(\mathbb{R})$, $A_2 \in M_{m_2 \times k}(\mathbb{R})$, $C \in M_{c \times k}(\mathbb{R})$, $B_1 \in M_{m_1 \times \ell}(\mathbb{R})$, $B_2 \in M_{m_2 \times \ell}(\mathbb{R})$, $D \in M_{c \times \ell}(\mathbb{R})$. Assume the following conditions hold:*

$$(1) \text{ rank } A_2 = \text{rank} \begin{pmatrix} A_1 \\ A_2 \end{pmatrix} = \text{rank} \begin{pmatrix} A_1 & B_1 \\ A_2 & B_2 \end{pmatrix},$$

$$(2) \text{ rank} \begin{pmatrix} A_2 \\ C \end{pmatrix} = \text{rank} \begin{pmatrix} A_2 & B_2 \\ C & D \end{pmatrix},$$

Then

$$\text{rank} \begin{pmatrix} A_1 \\ A_2 \\ C \end{pmatrix} = \text{rank} \begin{pmatrix} A_1 & B_1 \\ A_2 & B_2 \\ C & D \end{pmatrix}.$$

Proof. The assumptions (1) and (2) imply that there exist matrices $U \in M_{m_1 \times m_2}(\mathbb{R})$, $V \in M_{k \times \ell}(\mathbb{R})$, $W \in M_{k \times \ell}(\mathbb{R})$, such that

$$(2.14) \quad A_1 = UA_2, \quad B_1 = A_1V, \quad B_2 = A_2V, \quad B_2 = A_2W, \quad D = CW.$$

Using (2.14) we get

$$B_1 = A_1V = UA_2V = UB_2 = UA_2W = A_1W$$

and therefore

$$\begin{pmatrix} B_1 \\ B_2 \\ D \end{pmatrix} = \begin{pmatrix} A_1W \\ A_2W \\ CW \end{pmatrix} = \begin{pmatrix} A_1 \\ A_2 \\ C \end{pmatrix} W,$$

which completes the proof. $\square$

3. PROOF OF THE IMPLICATION (2) $\Rightarrow$ (1) AND THE MOREOVER PART OF THEOREM 1.1

Let $S_i := L(x^i)$, $T_i := L((x-t)^i)$ for $i \in [0; 2n]$ and $\mathcal{S} := (S_i)_{i \in [0; 2n]}$, $\mathcal{T} := (T_i)_{i \in [0; 2n]}$. Let $X_{\mathcal{T}}^i \equiv \text{row} \left((x_{\mathcal{T}}^{(i)})_j \right)_{j \in [p]}$, $\text{Dep}(X_{\mathcal{T}}^i)$, $\text{DepN}(X_{\mathcal{T}}^i)$ and $\text{DepN}_{\ell}(X_{\mathcal{T}}^i)$, $\ell = 0, 1$, be as in (2.4), (2.5) and (2.7), respectively. Write

$$(x_{\mathcal{T}}^{(i)})_j =: \begin{pmatrix} u_j^{(i)} \\ v_j^{(i)} \end{pmatrix} \in \begin{pmatrix} \mathbb{R}^{np} \\ \mathbb{R}^p \end{pmatrix} \quad \text{and} \quad \text{col}(T_{n+j})_{j \in [n]} =: \text{row}(k_i)_{i \in [p]} \in M_{np \times p}(\mathbb{R}), \quad k_i \in \mathbb{R}^{np}.$$

Note that in the formulation of Theorem 1.1, we have

$$\mathbf{y}_{ip+j} = (x_{\mathcal{T}}^{(i)})_j, \quad \mathbf{u}_{ip+j} = \begin{cases} u_j^{(i)}, & \text{for } i \in [0; n], \\ k_j, & \text{for } i = n+1, \end{cases}$$

$$\mathcal{A} = \bigcup_{i=1}^n \text{DepN}_1(X_{\mathcal{T}}^i),$$

$$\mathcal{B} = \left\{ j \in [p] : j \notin \text{Dep}(X_{\mathcal{T}}^n), k_j \in \text{span}\{u_1^{(1)}, u_2^{(1)}, \dots, u_p^{(n)}, k_1, \dots, k_{j-1}\} \right\}.$$

In the first step of the proof, we will build relation matrices that represent column relations of $M_{\mathcal{T}}(n)$. By Proposition 2.6, $M_S(n)$ is block-recursively generated and by Proposition 2.10, so is $M_{\mathcal{T}}(n)$. Let

$$(3.1) \quad \mathcal{D} := \{i \in [0; n] : \text{DepN}(X_{\mathcal{T}}^i) \neq \emptyset\}.$$

If $\mathcal{D} \neq \emptyset$, write

$$(3.2) \quad \mathcal{D} = \{z_1, \dots, z_s : 0 \leq z_1 < z_2 < \dots < z_s \leq n\}.$$

The following permutation organizes the coordinate indices according to the degree at which their columns first become dependent on preceding columns. We apply the same permutation within every block column $X_{\mathcal{T}}^i$. The indices in $[p] \setminus \text{Dep}(X_{\mathcal{T}}^{z_s})$ form the first group: their columns remain independent of preceding columns throughout $M_{\mathcal{T}}(n)$. The remaining indices are grouped according to the degree z_j at which a new dependence appears. Within each group, the subdivision into $\text{DepN}_0(X_{\mathcal{T}}^{z_j})$ and $\text{DepN}_1(X_{\mathcal{T}}^{z_j})$ distinguishes whether the dependence requires columns from $X_{\mathcal{T}}^0$ or can be expressed without them.

We arrange these groups in decreasing order of the degrees $z_s, \dots, z_1$. By recursive generation, columns whose dependence appeared at an earlier degree can be eliminated from relations at later degrees using their existing relations. Placing these groups farther to the right allows the new relations to be organized in triangular form. This is the structural reason for the ordering: it yields the triangular coefficient matrix $\check{H}_n$ whose invertibility is used in Claim 2.

Accordingly, let $Q \in M_p(\mathbb{R})$ be a permutation matrix which permutes the columns of the given matrix, when multiplied from the right, to the order

$$(3.3) \quad \begin{aligned} & [p] \setminus \text{Dep}(X_{\mathcal{T}}^{z_s}), \text{DepN}_0(X_{\mathcal{T}}^{z_s}), \text{DepN}_1(X_{\mathcal{T}}^{z_s}), \text{DepN}_0(X_{\mathcal{T}}^{z_{s-1}}), \text{DepN}_1(X_{\mathcal{T}}^{z_{s-1}}), \dots, \\ & \text{DepN}_0(X_{\mathcal{T}}^{z_2}), \text{DepN}_1(X_{\mathcal{T}}^{z_2}), \text{DepN}_0(X_{\mathcal{T}}^{z_1}), \text{DepN}_1(X_{\mathcal{T}}^{z_1}). \end{aligned}$$

Let

$$(3.4) \quad \begin{aligned} p_0 &:= p - \text{card Dep}(X_{\mathcal{T}}^{z_s}), \\ p_{j,\ell} &:= \text{card DepN}_{\ell}(X_{\mathcal{T}}^{z_{s-j+1}}) \text{ for } j \in [s] \text{ and } \ell = 0, 1, \\ p_j &:= \text{card DepN}(X_{\mathcal{T}}^{z_{s-j+1}}) = p_{j,0} + p_{j,1} \text{ for } j \in [s], \end{aligned}$$

where the last equality follows from Lemma 2.2.

If $\mathcal{D} = \emptyset$, we set

$$s := 0, \quad z_s \equiv z_0 := 0, \quad p_0 := p, \quad Q := I_p.$$

By Lemma 2.2 and Corollary 2.7, the sets $\text{DepN}_1(X_{\mathcal{T}}^{z_j})$, $j \in [s]$, are pairwise disjoint. Hence

$$\text{card } \mathcal{A} = \sum_{j=1}^s p_{j,1},$$

where the sum is understood to be zero if $s = 0$.

Claim 1. We have

$$\begin{aligned} \text{rank } M(n) &= \text{rank } M_{\mathcal{T}}(n) = (n+1)p_0 + \sum_{j=1}^s z_{s-j+1}p_j, \\ \text{rank row}(X_{\mathcal{T}}^i)_{i \in [n]} &= \text{rank } M_{\mathcal{T}}(n) - p_0 - \text{card } \mathcal{A}. \end{aligned}$$

Proof of Claim 1. By Proposition 2.8(5), $\text{rank } M(n) = \text{rank } M_{\mathcal{T}}(n)$. We compute the latter rank by counting the columns that are independent of their predecessors.

By Corollary 2.7, the sets $\text{Dep}(X_{\mathcal{T}}^i)$ are nested. Moreover, by the definition of $\mathcal{D}$, no new dependencies appear after degree z_s , so $\text{Dep}(X_{\mathcal{T}}^n) = \text{Dep}(X_{\mathcal{T}}^{z_s})$. Consequently, by the definition of p_0 , the set $[p] \setminus \text{Dep}(X_{\mathcal{T}}^n)$ has cardinality p_0 . Each of the p_0 indices outside $\text{Dep}(X_{\mathcal{T}}^n)$ therefore contributes $n + 1$ to the rank. If $r \in \text{DepN}(X_{\mathcal{T}}^{z_{s-j+1}})$, then, by the definition of DepN and the nesting of the sets $\text{Dep}(X_{\mathcal{T}}^i)$, $r \notin \text{Dep}(X_{\mathcal{T}}^i)$ precisely for $i \in [0; z_{s-j+1} - 1]$. Thus each of the p_j indices in this group contributes z_{s-j+1} . Since $[p] \setminus \text{Dep}(X_{\mathcal{T}}^n)$ together with the sets $\text{DepN}(X_{\mathcal{T}}^{z_{s-j+1}})$, $j \in [s]$, forms a partition of $[p]$, we obtain

$$\text{rank } M(n) = \text{rank } M_{\mathcal{T}}(n) = (n + 1)p_0 + \sum_{j=1}^s z_{s-j+1}p_j.$$

This also covers $\mathcal{D} = \emptyset$, when $s = 0$ and $p_0 = p$.

For the second identity, Corollary 2.7 and the definition of DepN_1 give

$$\text{Dep}_1(X_{\mathcal{T}}^i) = \text{Dep}(X_{\mathcal{T}}^{i-1}) \sqcup \text{DepN}_1(X_{\mathcal{T}}^i), \quad i \in [n].$$

Moreover, the sets $\text{DepN}_1(X_{\mathcal{T}}^i)$ are pairwise disjoint and their union is $\mathcal{A}$. Counting columns that are independent of their predecessors after omitting $X_{\mathcal{T}}^0$, we therefore obtain

$$\begin{aligned} \text{rank row}(X_{\mathcal{T}}^i)_{i \in [n]} &= \sum_{i=1}^n (p - \text{card } \text{Dep}_1(X_{\mathcal{T}}^i)) \\ &= \sum_{i=0}^{n-1} (p - \text{card } \text{Dep}(X_{\mathcal{T}}^i)) - \text{card } \mathcal{A} \\ &= \text{rank } M_{\mathcal{T}}(n) - (p - \text{card } \text{Dep}(X_{\mathcal{T}}^n)) - \text{card } \mathcal{A} \\ &= \text{rank } M_{\mathcal{T}}(n) - p_0 - \text{card } \mathcal{A}. \end{aligned}$$

In the third equality, we used $\text{rank } M_{\mathcal{T}}(n) = \sum_{i=0}^n (p - \text{card } \text{Dep}(X_{\mathcal{T}}^i))$, which counts the columns that are independent of their predecessors. $\blacksquare$

Let us define the permuted matrix

$$(3.5) \quad \text{row}(\check{X}_{\mathcal{T}}^i)_{i \in [0; n]} := \text{row}(X_{\mathcal{T}}^i Q)_{i \in [0; n]}.$$

Write

$$(3.6) \quad \check{X}_{\mathcal{T}}^i = \left(\underbrace{(\check{X}_{\mathcal{T}}^i)_{(0)}}_{p_0 \text{ columns}} \underbrace{(\check{X}_{\mathcal{T}}^i)_{(1,0)}}_{p_{1,0} \text{ columns}} \underbrace{(\check{X}_{\mathcal{T}}^i)_{(1,1)}}_{p_{1,1} \text{ columns}} \underbrace{(\check{X}_{\mathcal{T}}^i)_{(2,0)}}_{p_{2,0} \text{ columns}} \underbrace{(\check{X}_{\mathcal{T}}^i)_{(2,1)}}_{p_{2,1} \text{ columns}} \cdots \underbrace{(\check{X}_{\mathcal{T}}^i)_{(s,0)}}_{p_{s,0} \text{ columns}} \underbrace{(\check{X}_{\mathcal{T}}^i)_{(s,1)}}_{p_{s,1} \text{ columns}} \right).$$

Claim 2. There exist matrices $H_0, H_1, \dots, H_{n-1} \in M_{p \times (p-p_0)}(\mathbb{R})$, $\check{H}_{n,(0)} \in M_{p_0 \times (p-p_0)}(\mathbb{R})$ and an invertible matrix $\check{H}_n \in M_{p-p_0}(\mathbb{R})$ such that for

$$(3.7) \quad G := \text{col} \left(\text{col}(H_i)_{i \in [0; n-1]}, \check{H}_{n,(0)} \right),$$

we have

$$(3.8) \quad \text{row} \left((\check{X}_{\mathcal{T}}^n)_{(i,0)} (\check{X}_{\mathcal{T}}^n)_{(i,1)} \right)_{i \in [s]} \check{H}_n = \left(\text{row}(X_{\mathcal{T}}^i)_{i \in [0; n-1]} (\check{X}_{\mathcal{T}}^n)_{(0)} \right) G.$$

For $i \in [0; n-1]$ let $W_i := \bigcap_{j=0}^i \text{Ker } H_j$ and $s_i := \dim W_i$. Write $r_j := \sum_{\ell=1}^j p_{\ell}$ for $j \in [s]$. Then

$$(3.9) \quad \sum_{i=0}^{n-1} s_i \geq (n - z_s)(p - p_0) + \sum_{j=1}^{s-1} (z_{j+1} - z_j)(p - p_0 - r_{s-j}) + \text{card } \mathcal{A}.$$

Proof of Claim 2. Let $r_0 := 0$. By definition of the indices z_j , for $j \in [s]$ and $\ell = 0, 1$ there exist matrices

$$(3.10) \quad \begin{aligned} \check{B}_0^{(j,\ell)}, \check{B}_1^{(j,\ell)}, \dots, \check{B}_{z_j-1}^{(j,\ell)} &\in M_{p \times p_{s-j+1,\ell}}(\mathbb{R}), \quad \check{B}_{z_j,(0)}^{(j,\ell)} \in M_{p_0 \times p_{s-j+1,\ell}}(\mathbb{R}) \quad \text{and} \\ \check{B}_{z_j}^{(j,\ell)} &\in M_{(r_{s-j} + \sum_{i=0}^{\ell-1} p_{s-j+1,i}) \times p_{s-j+1,\ell}}(\mathbb{R}) \end{aligned}$$

(note that if $\ell = 0$, then $\sum_{i=0}^{\ell-1} p_{s-j+1,i} = 0$) such that

$$(3.11) \quad \begin{aligned} (\check{X}_{\mathcal{T}}^{z_j})_{(s-j+1,\ell)} &= \text{row}(\check{X}_{\mathcal{T}}^i)_{i \in [0; z_j-1]} \text{col}(\check{B}_i^{(j,\ell)})_{i \in [0; z_j-1]} + (\check{X}_{\mathcal{T}}^{z_j})_{(0)} \check{B}_{z_j,(0)}^{(j,\ell)} \\ &\quad + \left(\text{row}((\check{X}_{\mathcal{T}}^{z_j})_{(i,0)} (\check{X}_{\mathcal{T}}^{z_j})_{(i,1)})_{i \in [s-j]} \quad \text{row}((\check{X}_{\mathcal{T}}^{z_j})_{(s-j+1,i)})_{i \in [0;\ell-1]} \right) \check{B}_{z_j}^{(j,\ell)}. \end{aligned}$$

Note that if $\ell = 0$, then $\text{row}((\check{X}_{\mathcal{T}}^{z_j})_{(s-j+1,i)})_{i \in [0;\ell-1]}$ is an empty matrix. By the ordering of the columns (see (3.3)), note that $\check{B}_0^{(j,1)} = \mathbf{0}_{p \times p_{s-j+1,\ell}}$ for each $j \in [s]$. By Proposition 2.6, multiplying the relations in (3.11) by $\check{X}_{\mathcal{T}}^{n-z_j}$, we obtain for $j \in [s]$ and $\ell = 0, 1$ the relations

$$\begin{aligned} &\left(\text{row}((\check{X}_{\mathcal{T}}^n)_{(i,0)} (\check{X}_{\mathcal{T}}^n)_{(i,1)})_{i \in [s-j]} \quad \text{row}((\check{X}_{\mathcal{T}}^n)_{(s-j+1,i)})_{i \in [0;\ell-1]} \quad (\check{X}_{\mathcal{T}}^n)_{(s-j+1,\ell)} \right) \begin{pmatrix} -\check{B}_{z_j}^{(j,\ell)} \\ I_{p_{s-j+1,\ell}} \end{pmatrix} \\ &= \text{row}(\check{X}_{\mathcal{T}}^{n-z_j+i})_{i \in [0; z_j-1]} \text{col}(\check{B}_i^{(j,\ell)})_{i \in [0; z_j-1]} + (\check{X}_{\mathcal{T}}^n)_{(0)} \check{B}_{z_j,(0)}^{(j,\ell)} \end{aligned}$$

of $M_{\mathcal{T}}(n)$, or equivalently for $j \in [s]$ we have

$$(3.12) \quad \text{row}((\check{X}_{\mathcal{T}}^n)_{(i,0)} (\check{X}_{\mathcal{T}}^n)_{(i,1)})_{i \in [s]} \check{H}_n^{(j)} = \sum_{i=0}^{n-1} \check{X}_{\mathcal{T}}^i \check{H}_i^{(j)} + (\check{X}_{\mathcal{T}}^n)_{(0)} \check{H}_{n,(0)}^{(j)},$$

where

$$\begin{aligned} \check{H}_i^{(j)} &:= \mathbf{0}_{p \times (p-p_0)} \quad \text{for } i \in [0; n - z_j - 1], \\ \check{H}_i^{(j)} &:= (\mathbf{0}_{p \times r_{s-j}} \quad \check{B}_{i-n+z_j}^{(j,0)} \quad \check{B}_{i-n+z_j}^{(j,1)} \quad \mathbf{0}_{p \times (p-p_0-r_{s-j+1})}) \in M_{p \times (p-p_0)}(\mathbb{R}) \quad \text{for } i \in [n - z_j; n - 1], \\ \check{H}_{n,(0)}^{(j)} &:= (\mathbf{0}_{p_0 \times r_{s-j}} \quad \check{B}_{z_j,(0)}^{(j,0)} \quad \check{B}_{z_j,(0)}^{(j,1)} \quad \mathbf{0}_{p_0 \times (p-p_0-r_{s-j+1})}) \in M_{p_0 \times (p-p_0)}(\mathbb{R}), \\ \check{H}_n^{(j)} &:= \begin{pmatrix} \mathbf{0}_{r_{s-j}} & \begin{pmatrix} -\check{B}_{z_j}^{(j,0)} \\ I_{p_{s-j+1,0}} \end{pmatrix} & -\check{B}_{z_j}^{(j,1)} \\ \mathbf{0} & \mathbf{0}_{p_{s-j+1,1} \times p_{s-j+1,0}} & I_{p_{s-j+1,1}} \end{pmatrix} \oplus \mathbf{0}_{p-p_0-r_{s-j+1}} \in M_{p-p_0}(\mathbb{R}). \end{aligned}$$

Let

$$\begin{aligned} \check{B}_{z_j}^{(j,0)} &:= \text{col}(\check{B}_{z_j,i}^{(j,0)})_{i \in [s-j]} \quad \text{and} \\ \check{B}_{z_j}^{(j,1)} &:= \text{col}(\text{col}(\check{B}_{z_j,i}^{(j,1)})_{i \in [s-j]}, \check{B}_{z_j,s-j+1,0}^{(j,1)}) \quad \text{for } j \in [s], \end{aligned}$$

where $\check{B}_{z_j,i}^{(j,\ell)} \in M_{p_i \times p_{s-j+1,\ell}}(\mathbb{R})$, $\check{B}_{z_j,s-j+1,0}^{(j,1)} \in M_{p_{s-j+1,0} \times p_{s-j+1,1}}(\mathbb{R})$ for $j \in [s]$, $i \in [s-j]$ and $\ell = 0, 1$. Summing all block column relations (3.12) for $j \in [s]$, we get a block column relation

$$(3.13) \quad \text{row}((\check{X}_{\mathcal{T}}^n)_{(i,0)} (\check{X}_{\mathcal{T}}^n)_{(i,1)})_{i \in [s]} \check{H}_n = \sum_{i=0}^{n-1} \check{X}_{\mathcal{T}}^i \check{H}_i + (\check{X}_{\mathcal{T}}^n)_{(0)} \check{H}_{n,(0)},$$

where

$$\begin{aligned}
 \check{H}_i &:= \sum_{j=1}^s \check{H}_i^{(j)} \\
 (3.14) \quad &= \left\{ \begin{array}{ll} \mathbf{0}_{p \times (p-p_0)}, & \text{for } i \in [0; n - z_s - 1], \\ \left(\underbrace{\check{B}_{i-n+z_s}^{(s,0)}}_{p_{1,0} \text{ columns}} \quad \underbrace{\mathbf{0}}_{p-p_0-p_{1,0} \text{ columns}} \right), & \text{for } i = n - z_s, \\ \left(\underbrace{\check{B}_{i-n+z_s}^{(s,0)}}_{p_{1,0} \text{ columns}} \quad \underbrace{\check{B}_{i-n+z_s}^{(s,1)}}_{p_{1,1} \text{ columns}} \quad \underbrace{\mathbf{0}}_{p-p_0-p_1 \text{ columns}} \right), & \text{for } i \in [n - z_s + 1; n - z_{s-1} - 1], \\ \left(\underbrace{\check{B}_{i-n+z_s}^{(s,0)}}_{p_{1,0} \text{ columns}} \quad \underbrace{\check{B}_{i-n+z_s}^{(s,1)}}_{p_{1,1} \text{ columns}} \quad \underbrace{\check{B}_{i-n+z_{s-1}}^{(s-1,0)}}_{p_{2,0} \text{ columns}} \quad \underbrace{\mathbf{0}}_{p-p_0-p_1-p_{2,0} \text{ columns}} \right), & \text{for } i = n - z_{s-1}, \\ \left(\underbrace{\check{B}_{i-n+z_s}^{(s,0)}}_{p_{1,0} \text{ columns}} \quad \underbrace{\check{B}_{i-n+z_s}^{(s,1)}}_{p_{1,1} \text{ columns}} \quad \underbrace{\check{B}_{i-n+z_{s-1}}^{(s-1,0)}}_{p_{2,0} \text{ columns}} \quad \underbrace{\check{B}_{i-n+z_{s-1}}^{(s-1,1)}}_{p_{2,1} \text{ columns}} \quad \underbrace{\mathbf{0}}_{p-p_0-p_1-p_2 \text{ columns}} \right), & \\ \vdots & \\ \left(\text{row} \left(\underbrace{\check{B}_{i-n+z_{s-j+1}}^{(s-j+1,0)}}_{p_{j,0} \text{ columns}} \quad \underbrace{\check{B}_{i-n+z_{s-j+1}}^{(s-j+1,1)}}_{p_{j,1} \text{ columns}} \right)_{j \in [s-1]} \quad \underbrace{\check{B}_{i-n+z_1}^{(1,0)}}_{p_{s,0} \text{ columns}} \quad \underbrace{\mathbf{0}}_{p-p_0-r_{s-1}-p_{s,0} \text{ columns}} \right), & \\ \vdots & \\ \text{row} \left(\underbrace{\check{B}_{i-n+z_{s-j+1}}^{(s-j+1,0)}}_{p_{j,0} \text{ columns}} \quad \underbrace{\check{B}_{i-n+z_{s-j+1}}^{(s-j+1,1)}}_{p_{j,1} \text{ columns}} \right)_{j \in [s]}, & \text{for } i \in [n - z_1 + 1; n - 1], \end{array} \right.
 \end{aligned}$$

and

$$\begin{aligned}
 \check{H}_{n,(0)} &= \text{row} \left(\underbrace{\check{B}_{z_{s-j+1},(0)}^{(s-j+1,0)}}_{p_{j,0} \text{ columns}} \quad \underbrace{\check{B}_{z_{s-j+1},(0)}^{(s-j+1,1)}}_{p_{j,1} \text{ columns}} \right)_{j \in [s]}, \\
 \check{H}_n &= \begin{pmatrix} D_1 & -\text{row} \left(\check{B}_{z_{s-1},1}^{(s-1,\ell)} \right)_{\ell=0,1} & -\text{row} \left(\check{B}_{z_{s-2},1}^{(s-2,\ell)} \right)_{\ell=0,1} & \cdots & -\text{row} \left(\check{B}_{z_1,1}^{(1,\ell)} \right)_{\ell=0,1} \\ \mathbf{0} & D_2 & -\text{row} \left(\check{B}_{z_{s-2},2}^{(s-2,\ell)} \right)_{\ell=0,1} & \cdots & -\text{row} \left(\check{B}_{z_1,2}^{(1,\ell)} \right)_{\ell=0,1} \\ \vdots & \ddots & & \ddots & \vdots \\ \vdots & & & \ddots & -\text{row} \left(\check{B}_{z_1,s-1}^{(1,\ell)} \right)_{\ell=0,1} \\ \mathbf{0} & \cdots & \cdots & \mathbf{0} & D_s \end{pmatrix},
 \end{aligned}$$

where $D_j := \begin{pmatrix} I_{p_{j,0}} & -\check{B}_{z_{s-j+1},j,0}^{(s-j+1,1)} \\ \mathbf{0} & I_{p_{j,1}} \end{pmatrix}$ for $j \in [s]$. Note that $\check{H}_n$ is invertible. It follows from (3.13) that

$$\begin{aligned} \text{row}((\check{X}_{\mathcal{T}}^n)_{(i,0)} (\check{X}_{\mathcal{T}}^n)_{(i,1)})_{i \in [s]} \check{H}_n &= \sum_{i=0}^{n-1} X_{\mathcal{T}}^i H_i + (\check{X}_{\mathcal{T}}^n)_{(0)} \check{H}_{n,(0)} \\ &= \left(\text{row}(X_{\mathcal{T}}^i)_{i \in [0;n-1]} \quad (\check{X}_{\mathcal{T}}^n)_{(0)} \right) G. \end{aligned}$$

where $H_i := Q\check{H}_i$ for $i \in [0; n-1]$ and G is as in (3.7). This proves (3.8). It remains to prove (3.9). For $i \in [0; n-1]$ let s_i be defined as in the statement of Claim 2. Using (3.14) we have

$$(3.15) \quad s_i = \begin{cases} p - p_0, & \text{for } i \in [0; n - z_s - 1], \\ \text{at least } p - p_0 - p_{1,0}, & \text{for } i = n - z_s, \\ \text{at least } p - p_0 - p_1, & \text{for } i \in [n - z_s + 1; n - z_{s-1} - 1], \\ \text{at least } p - p_0 - p_1 - p_{2,0}, & \text{for } i = n - z_{s-1}, \\ \text{at least } p - p_0 - p_1 - p_2, & \text{for } i \in [n - z_{s-1} + 1; n - z_{s-2} - 1], \\ \vdots \\ \text{at least } p - p_0 - r_{s-2} - p_{s-1,0}, & \text{for } i = n - z_2, \\ \text{at least } p - p_0 - r_{s-1}, & \text{for } i \in [n - z_2 + 1; n - z_1 - 1], \\ \text{at least } p - p_0 - r_{s-1} - p_{s,0}, & \text{for } i = n - z_1. \end{cases}$$

It follows from (3.15) that

$$\begin{aligned} \sum_{i=0}^{n-1} s_i &\geq (n - z_s)(p - p_0) + \sum_{j=1}^s (p - p_0 - r_{s-j} - p_{s-j+1,0}) \\ &\quad + \sum_{j=1}^{s-1} (z_{j+1} - z_j - 1)(p - p_0 - r_{s-j}) \\ &= (n - z_s)(p - p_0) + \sum_{j=1}^{s-1} (z_{j+1} - z_j)(p - p_0 - r_{s-j}) + p - p_0 - \underbrace{\sum_{j=1}^s p_{s-j+1,0}}_{=\sum_{j=1}^s p_{j,1} = \text{card } \mathcal{A}}, \end{aligned}$$

which is (3.9). This concludes the proof of Claim 2. ■

The continuation of the proof will be similar to the proof of implication (2) $\Rightarrow$ (1) in [ZZ26, Theorem 1.1], while accounting for the column relations of the matrix $M_{\mathcal{T}}(n)$, which are transferred to the matrix K in (3.16) by recursive generation. Write

$$(3.16) \quad X_{\mathcal{T}}^0 =: \begin{pmatrix} \mathcal{H}_0 \\ T_n \end{pmatrix} \quad \text{and} \quad \text{row}(X_{\mathcal{T}}^i)_{i \in [n]} =: \begin{pmatrix} \mathcal{H} \\ K^T \end{pmatrix} \in \begin{pmatrix} \mathbb{S}_{np}(\mathbb{R}) \\ M_{p \times np}(\mathbb{R}) \end{pmatrix}.$$

Let

$$(3.17) \quad k := \text{rank} \begin{pmatrix} \mathcal{H} \\ K^T \end{pmatrix} - \text{rank } \mathcal{H} = \text{rank} \begin{pmatrix} \mathcal{H} & K \end{pmatrix} - \text{rank } \mathcal{H}.$$

Let $\mathcal{B}$ be as in the statement of Theorem 1.1. Note that

$$\mathcal{B} \subseteq [p] \setminus \text{Dep}(X_{\mathcal{T}}^{z_s}) \quad \text{and} \quad \text{card}([p] \setminus (\mathcal{B} \sqcup \text{Dep}(X_{\mathcal{T}}^{z_s}))) = k.$$

(Here $\sqcup$ stands for the disjoint union.) Let $P_1 \in M_{p_0}(\mathbb{R})$ be a permutation matrix, which permutes the columns in $[p] \setminus \text{Dep}(X_{\mathcal{T}}^{zs})$ to the order C_1, C_2 , where

$$[p] \setminus (\mathcal{B} \sqcup \text{Dep}(X_{\mathcal{T}}^{zs})) \subseteq C_1 \quad \text{and} \quad k \leq \text{card } C_1 := p_0 - m + \text{card } \mathcal{A}.$$

By the assumption $\text{card } \mathcal{A} \leq m \leq \text{card}(\mathcal{A} \cup \mathcal{B})$, it follows that $0 \leq \text{card } C_1 \leq p_0$. Recall that Q permutes the columns of the given matrix, when multiplied from the right, to the order (3.3). Then for the permutation matrix $P := Q(P_1 \oplus I_{p-p_0}) \in M_p(\mathbb{R})$ and

$$(3.18) \quad \hat{K} \equiv \left(\underbrace{\hat{K}_1}_{p_0-m+\text{card } \mathcal{A} \text{ columns}} \quad \underbrace{\hat{K}_2}_{m-\text{card } \mathcal{A} \text{ columns}} \quad \underbrace{\hat{K}_3}_{p-p_0 \text{ columns}} \right) := KP,$$

we have that

$$(3.19) \quad \text{rank} \begin{pmatrix} \mathcal{H} & \hat{K}_1 \end{pmatrix} = \text{rank} \begin{pmatrix} \mathcal{H} & \hat{K} \end{pmatrix} = \text{rank } \mathcal{H} + k.$$

By (3.8) and recursive generation (see Proposition 2.6), it follows that

$$(3.20) \quad \hat{K}_3 = \begin{pmatrix} \mathcal{H} & \hat{K}_1 & \hat{K}_2 \end{pmatrix} \hat{G},$$

where

$$(3.21) \quad \hat{G} := (I_{np} \oplus P_1^T) G \check{H}_n^{-1},$$

or equivalently

$$(3.22) \quad \begin{pmatrix} \mathcal{H} & \hat{K}_1 & \hat{K}_2 & \hat{K}_3 \end{pmatrix} \begin{pmatrix} -\hat{G} \\ I_{p-p_0} \end{pmatrix} = \mathbf{0}_{np \times (p-p_0)}.$$

By (3.19), it follows that

$$(3.23) \quad \hat{K}_2 = \begin{pmatrix} \mathcal{H} & \hat{K}_1 \end{pmatrix} J$$

for some real matrix $J \in M_{(np+p_0-m+\text{card } \mathcal{A}) \times (m-\text{card } \mathcal{A})}(\mathbb{R})$, or equivalently

$$(3.24) \quad \begin{pmatrix} \mathcal{H} & \hat{K}_1 & \hat{K}_2 \end{pmatrix} \begin{pmatrix} -J \\ I_{m-\text{card } \mathcal{A}} \end{pmatrix} = \mathbf{0}_{np \times (m-\text{card } \mathcal{A})}$$

Claim 3. There exists $\hat{Z} \in \mathbb{S}_p(\mathbb{R})$ such that

$$(3.25) \quad \text{rank} \begin{pmatrix} \mathcal{H} & \hat{K} \\ \hat{K}^T & \hat{Z} \end{pmatrix} = \text{rank} \begin{pmatrix} \mathcal{H} \\ \hat{K}^T \end{pmatrix} + p_0 - m + \text{card } \mathcal{A}$$

and

$$(3.26) \quad \begin{pmatrix} \mathcal{H} & \hat{K} \\ \hat{K}^T & \hat{Z} \end{pmatrix} \begin{pmatrix} -J_1 & -\hat{G}_1 \\ -J_2 & -\hat{G}_2 \\ I_{m-\text{card } \mathcal{A}} & -\hat{G}_3 \\ \mathbf{0} & I_{p-p_0} \end{pmatrix} = \mathbf{0}_{(n+1)p \times (m-\text{card } \mathcal{A} + p - p_0)},$$

where

$$J =: \begin{pmatrix} J_1 \\ J_2 \end{pmatrix} \in \begin{pmatrix} M_{np \times (m-\text{card } \mathcal{A})}(\mathbb{R}) \\ M_{(p_0-m+\text{card } \mathcal{A}) \times (m-\text{card } \mathcal{A})}(\mathbb{R}) \end{pmatrix},$$

$$\hat{G} =: \begin{pmatrix} \hat{G}_1 \\ \hat{G}_2 \\ \hat{G}_3 \end{pmatrix} \in \begin{pmatrix} M_{np \times (p-p_0)}(\mathbb{R}) \\ M_{(p_0-m+\text{card } \mathcal{A}) \times (p-p_0)}(\mathbb{R}) \\ M_{(m-\text{card } \mathcal{A}) \times (p-p_0)}(\mathbb{R}) \end{pmatrix}.$$

Proof of Claim 3. By (3.22) and (3.24), to prove (3.26) it suffices to establish the equality

$$(3.27) \quad \begin{pmatrix} \widehat{K}^T & \widehat{Z} \end{pmatrix} \begin{pmatrix} -J_1 & -\widehat{G}_1 \\ -J_2 & -\widehat{G}_2 \\ I_{m-\text{card } \mathcal{A}} & -\widehat{G}_3 \\ \mathbf{0} & I_{p-p_0} \end{pmatrix} = \mathbf{0}_{p \times (m-\text{card } \mathcal{A} + p - p_0)}.$$

Let us write

$$(3.28) \quad \widehat{Z} := \begin{pmatrix} \widehat{Z}_1 & \widehat{Z}_2 & \widehat{Z}_3 \\ \widehat{Z}_2^T & \widehat{Z}_4 & \widehat{Z}_5 \\ \widehat{Z}_3^T & \widehat{Z}_5^T & \widehat{Z}_6 \end{pmatrix},$$

where $\widehat{Z}_1 \in M_{p_0-m+\text{card } \mathcal{A}}(\mathbb{R})$, $\widehat{Z}_2 \in M_{(p_0-m+\text{card } \mathcal{A}) \times (m-\text{card } \mathcal{A})}(\mathbb{R})$, $\widehat{Z}_3 \in M_{(p_0-m+\text{card } \mathcal{A}) \times (p-p_0)}(\mathbb{R})$, $\widehat{Z}_4 \in M_{m-\text{card } \mathcal{A}}(\mathbb{R})$, $\widehat{Z}_5 \in M_{(m-\text{card } \mathcal{A}) \times (p-p_0)}(\mathbb{R})$ and $\widehat{Z}_6 \in M_{p-p_0}(\mathbb{R})$. In this notation, (3.27) becomes

$$\begin{pmatrix} \widehat{K}_1^T & \widehat{Z}_1 & \widehat{Z}_2 & \widehat{Z}_3 \\ \widehat{K}_2^T & \widehat{Z}_2^T & \widehat{Z}_4 & \widehat{Z}_5 \\ \widehat{K}_3^T & \widehat{Z}_3^T & \widehat{Z}_5^T & \widehat{Z}_6 \end{pmatrix} \begin{pmatrix} -J_1 & -\widehat{G}_1 \\ -J_2 & -\widehat{G}_2 \\ I_{m-\text{card } \mathcal{A}} & -\widehat{G}_3 \\ \mathbf{0} & I_{p-p_0} \end{pmatrix} = \mathbf{0}_{p \times (m-\text{card } \mathcal{A} + p - p_0)}.$$

We choose $\widehat{Z}_1 \in \mathbb{S}_{p_0-m+\text{card } \mathcal{A}}(\mathbb{R})$ such that

$$(3.29) \quad \text{rank} \begin{pmatrix} \mathcal{H} & \widehat{K}_1 \\ \widehat{K}_1^T & \widehat{Z}_1 \end{pmatrix} = \text{rank} \begin{pmatrix} \mathcal{H} \\ \widehat{K}_1^T \end{pmatrix} + p_0 - m + \text{card } \mathcal{A}$$

and define

$$(3.30) \quad \begin{aligned} \widehat{Z}_2 &:= \begin{pmatrix} \widehat{K}_1^T & \widehat{Z}_1 \end{pmatrix} J, & \widehat{Z}_3 &:= \begin{pmatrix} \widehat{K}_1^T & \widehat{Z}_1 & \widehat{Z}_2 \end{pmatrix} \widehat{G}, \\ \widehat{Z}_4 &:= \begin{pmatrix} \widehat{K}_2^T & \widehat{Z}_2^T \end{pmatrix} J, & \widehat{Z}_5 &:= \begin{pmatrix} \widehat{K}_2^T & \widehat{Z}_2^T & \widehat{Z}_4 \end{pmatrix} \widehat{G}, \\ & & \widehat{Z}_6 &:= \begin{pmatrix} \widehat{K}_3^T & \widehat{Z}_3^T & \widehat{Z}_5^T \end{pmatrix} \widehat{G}. \end{aligned}$$

Once we prove (3.27), (3.29) will also imply (3.25).

Let us check that $\widehat{Z}$ is symmetric. Since $\widehat{Z}_1 \in \mathbb{S}_{p_0-m+\text{card } \mathcal{A}}(\mathbb{R})$, we only need to show that $\widehat{Z}_4 \in \mathbb{S}_{m-\text{card } \mathcal{A}}(\mathbb{R})$ and $\widehat{Z}_6 \in \mathbb{S}_{p-p_0}(\mathbb{R})$. But this follows by the following computations:

$$\begin{aligned} \underbrace{\widehat{Z}_4}_{(3.30)} &= \begin{pmatrix} \widehat{K}_2^T & \widehat{Z}_2^T \end{pmatrix} J = \begin{pmatrix} \widehat{K}_2 \\ \widehat{Z}_2 \end{pmatrix}^T J \underbrace{=}_{(3.23), (3.30)} \left(\begin{pmatrix} \mathcal{H} & \widehat{K}_1 \\ \widehat{K}_1^T & \widehat{Z}_1 \end{pmatrix} J \right)^T J = J^T \begin{pmatrix} \mathcal{H} & \widehat{K}_1 \\ \widehat{K}_1^T & \widehat{Z}_1 \end{pmatrix}^T J, \\ \underbrace{\widehat{Z}_6}_{(3.30)} &= \begin{pmatrix} \widehat{K}_3^T & \widehat{Z}_3^T & \widehat{Z}_5^T \end{pmatrix} \widehat{G} = \begin{pmatrix} \widehat{K}_3 \\ \widehat{Z}_3 \\ \widehat{Z}_5 \end{pmatrix}^T \widehat{G} \underbrace{=}_{(3.20), (3.30)} \left(\begin{pmatrix} \mathcal{H} & \widehat{K}_1 & \widehat{K}_2 \\ \widehat{K}_1^T & \widehat{Z}_1 & \widehat{Z}_2 \\ \widehat{K}_2^T & \widehat{Z}_2^T & \widehat{Z}_4 \end{pmatrix} \widehat{G} \right)^T \widehat{G} \\ &= \widehat{G}^T \begin{pmatrix} \mathcal{H} & \widehat{K}_1 & \widehat{K}_2 \\ \widehat{K}_1^T & \widehat{Z}_1 & \widehat{Z}_2 \\ \widehat{K}_2^T & \widehat{Z}_2^T & \widehat{Z}_4 \end{pmatrix} \widehat{G}. \end{aligned}$$

We now observe the following:

$$\begin{aligned}
 \widehat{Z}_5^T &\stackrel{(3.30)}{=} \left(\left(\widehat{K}_2^T \quad \widehat{Z}_2^T \quad \widehat{Z}_4 \right) \widehat{G} \right)^T = \widehat{G}^T \begin{pmatrix} \widehat{K}_2 \\ \widehat{Z}_2 \\ \widehat{Z}_4 \end{pmatrix} \stackrel{(3.23), (3.30)}{=} \widehat{G}^T \begin{pmatrix} \mathcal{H} & \widehat{K}_1 \\ \widehat{K}_1^T & \widehat{Z}_1 \\ \widehat{K}_2^T & \widehat{Z}_2^T \end{pmatrix} J \\
 &= \left(\left(\begin{pmatrix} \mathcal{H} & \widehat{K}_1 & \widehat{K}_2 \\ \widehat{K}_1^T & \widehat{Z}_1 & \widehat{Z}_2 \end{pmatrix} \widehat{G} \right)^T J \right) \stackrel{(3.20), (3.30)}{=} \begin{pmatrix} \widehat{K}_3 \\ \widehat{Z}_3 \end{pmatrix}^T J = \begin{pmatrix} \widehat{K}_3^T & \widehat{Z}_3^T \end{pmatrix} J.
 \end{aligned}
 \tag{3.31}$$

From the definitions of $\widehat{Z}_i, i \in [2; 6]$, and the equality (3.31), it is clear that the matrix $\widehat{Z}$ satisfies (3.27). This proves Claim 3. $\blacksquare$

Define the vectors $c_1, c_2, \dots, c_{m-\text{card } \mathcal{A}+p-p_0}$ by

$$C \equiv \text{row}(c_i)_{i \in [m-\text{card } \mathcal{A}+p-p_0]} := (I_{np} \oplus P) \begin{pmatrix} -J_1 & -\widehat{G}_1 \\ -J_2 & -\widehat{G}_2 \\ I_{m-\text{card } \mathcal{A}} & -\widehat{G}_3 \\ \mathbf{0} & I_{p-p_0} \end{pmatrix},$$

Note that $c_1, c_2, \dots, c_{m-\text{card } \mathcal{A}+p-p_0}$ are linearly independent. Next, define the matrix Z by

$$Z := P \widehat{Z} P^T \in \mathbb{S}_p(\mathbb{R}).$$

Claim 4. $c_1, c_2, \dots, c_{m-\text{card } \mathcal{A}+p-p_0} \in \text{Ker} \begin{pmatrix} \mathcal{H} & K \\ K^T & Z \end{pmatrix}.$

Proof of Claim 4. Recall from (3.18) that $\widehat{K} = KP$. We have

$$\begin{aligned}
 \begin{pmatrix} \mathcal{H} & K \\ K^T & Z \end{pmatrix} C &= \begin{pmatrix} \mathcal{H} & \widehat{K} P^T \\ P \widehat{K}^T & P \widehat{Z} P^T \end{pmatrix} (I_{np} \oplus P) \begin{pmatrix} -J_1 & -\widehat{G}_1 \\ -J_2 & -\widehat{G}_2 \\ I_{m-\text{card } \mathcal{A}} & -\widehat{G}_3 \\ \mathbf{0} & I_{p-p_0} \end{pmatrix} \\
 &= (I_{np} \oplus P) \begin{pmatrix} \mathcal{H} & \widehat{K} \\ \widehat{K}^T & \widehat{Z} \end{pmatrix} (I_{np} \oplus P^T) (I_{np} \oplus P) \begin{pmatrix} -J_1 & -\widehat{G}_1 \\ -J_2 & -\widehat{G}_2 \\ I_{m-\text{card } \mathcal{A}} & -\widehat{G}_3 \\ \mathbf{0} & I_{p-p_0} \end{pmatrix} \\
 &= (I_{np} \oplus P) \begin{pmatrix} \mathcal{H} & \widehat{K} \\ \widehat{K}^T & \widehat{Z} \end{pmatrix} \begin{pmatrix} -J_1 & -\widehat{G}_1 \\ -J_2 & -\widehat{G}_2 \\ I_{m-\text{card } \mathcal{A}} & -\widehat{G}_3 \\ \mathbf{0} & I_{p-p_0} \end{pmatrix} \\
 &\stackrel{(3.26)}{=} (I_{np} \oplus P) \mathbf{0}_{(n+1)p \times (m-\text{card } \mathcal{A}+p-p_0)} = \mathbf{0}_{(n+1)p \times (m-\text{card } \mathcal{A}+p-p_0)}.
 \end{aligned}$$

This proves Claim 4. $\blacksquare$

Define

$$\widehat{T}_n \equiv \left(\underbrace{\widehat{T}_{n,1}}_{p_0-m+\text{card } \mathcal{A} \text{ columns}} \quad \underbrace{\widehat{T}_{n,2}}_{m-\text{card } \mathcal{A} \text{ columns}} \quad \underbrace{\widehat{T}_{n,3}}_{p-p_0 \text{ columns}} \right) := T_n P = T_n Q (P_1 \oplus I_{p-p_0}).
 \tag{3.32}$$

Claim 5. Recall the notation in (3.16) and recall from (3.18) that $\widehat{K} = KP = \text{row}(\widehat{K}_1, \widehat{K}_2, \widehat{K}_3)$. We have

$$(3.33) \quad \text{rank} \begin{pmatrix} \mathcal{H}_0 & \mathcal{H} & \widehat{K} \\ \widehat{T}_n^T & \widehat{K}^T & \widehat{Z} \end{pmatrix} = \text{rank} \begin{pmatrix} \mathcal{H}_0 & \mathcal{H} \\ \widehat{T}_n^T & \widehat{K}^T \end{pmatrix} = \text{rank } M_{\mathcal{T}}(n).$$

Proof of Claim 5. Since

$$\begin{pmatrix} \mathcal{H}_0 & \mathcal{H} \\ \widehat{T}_n^T & \widehat{K}^T \end{pmatrix} = (I_{np} \oplus P^T) \begin{pmatrix} \mathcal{H}_0 & \mathcal{H} \\ T_n & K^T \end{pmatrix} = (I_{np} \oplus P^T) M_{\mathcal{T}}(n),$$

the nontrivial equality in (3.33) is the first one. Observe that

$$(3.34) \quad \begin{aligned} & \begin{pmatrix} \mathcal{H}_0^T & \widehat{T}_n \\ \mathcal{H} & \widehat{K} \end{pmatrix} \\ &= M_{\mathcal{T}}(n)(I_{np} \oplus P) \\ &= \begin{pmatrix} \text{row}(X_{\mathcal{T}}^i)_{i \in [0;n-1]} & \check{X}_{\mathcal{T}}^n(P_1 \oplus I_{p-p_0}) \end{pmatrix} \\ &= \begin{pmatrix} \text{row}(X_{\mathcal{T}}^i)_{i \in [0;n-1]} & (\check{X}_{\mathcal{T}}^n)_{(0)} P_1 & \text{row}((\check{X}_{\mathcal{T}}^n)_{(i,0)} (\check{X}_{\mathcal{T}}^n)_{(i,1)})_{i \in [s]} \end{pmatrix} \\ &= \underbrace{\begin{pmatrix} \text{row}(X_{\mathcal{T}}^i)_{i \in [0;n-1]} & (\check{X}_{\mathcal{T}}^n)_{(0)} \end{pmatrix} (I_{np} \oplus P_1)}_{= \begin{pmatrix} \mathcal{H}_0^T & \widehat{T}_{n,1} & \widehat{T}_{n,2} \\ \mathcal{H} & \widehat{K}_1 & \widehat{K}_2 \end{pmatrix}} \underbrace{\text{row}((\check{X}_{\mathcal{T}}^n)_{(i,0)} (\check{X}_{\mathcal{T}}^n)_{(i,1)})_{i \in [s]}}_{= \begin{pmatrix} \widehat{T}_{n,3} \\ \widehat{K}_3 \end{pmatrix}}. \end{aligned}$$

By (3.8), (3.21) and (3.34), it follows that

$$(3.35) \quad \widehat{T}_{n,3} = \begin{pmatrix} \mathcal{H}_0^T & \widehat{T}_{n,1} & \widehat{T}_{n,2} \end{pmatrix} \widehat{G}.$$

We have

$$(3.36) \quad \begin{aligned} & \text{rank} \begin{pmatrix} \mathcal{H}_0 & \mathcal{H} & \widehat{K} \\ \widehat{T}_n^T & \widehat{K}^T & \widehat{Z} \end{pmatrix} \underbrace{=}_{(*)^T} \text{rank} \begin{pmatrix} \mathcal{H}_0^T & \widehat{T}_n \\ \mathcal{H} & \widehat{K} \\ \widehat{K}^T & \widehat{Z} \end{pmatrix} \\ &= \text{rank} \begin{pmatrix} \mathcal{H}_0^T & \widehat{T}_{n,1} & \widehat{T}_{n,2} & \widehat{T}_{n,3} \\ \mathcal{H} & \widehat{K}_1 & \widehat{K}_2 & \widehat{K}_3 \\ \widehat{K}_1^T & \widehat{Z}_1 & \widehat{Z}_2 & \widehat{Z}_3 \\ \widehat{K}_2^T & \widehat{Z}_2^T & \widehat{Z}_4 & \widehat{Z}_5 \\ \widehat{K}_3^T & \widehat{Z}_3^T & \widehat{Z}_5^T & \widehat{Z}_6 \end{pmatrix} \underbrace{=}_{\substack{(3.20), \\ (3.30), \\ (3.35)}} \text{rank} \begin{pmatrix} \mathcal{H}_0^T & \widehat{T}_{n,1} & \widehat{T}_{n,2} \\ \mathcal{H} & \widehat{K}_1 & \widehat{K}_2 \\ \widehat{K}_1^T & \widehat{Z}_1 & \widehat{Z}_2 \\ \widehat{K}_2^T & \widehat{Z}_2^T & \widehat{Z}_4 \\ \widehat{K}_3^T & \widehat{Z}_3^T & \widehat{Z}_5^T \end{pmatrix} \\ &\underbrace{=}_{(*)^T} \text{rank} \begin{pmatrix} \mathcal{H}_0 & \mathcal{H} & \widehat{K}_1 & \widehat{K}_2 & \widehat{K}_3 \\ \widehat{T}_{n,1}^T & \widehat{K}_1^T & \widehat{Z}_1 & \widehat{Z}_2 & \widehat{Z}_3 \\ \widehat{T}_{n,2}^T & \widehat{K}_2^T & \widehat{Z}_2^T & \widehat{Z}_4 & \widehat{Z}_5 \end{pmatrix} \underbrace{=}_{\substack{(3.20), \\ (3.23), \\ (3.30)}} \text{rank} \begin{pmatrix} \mathcal{H}_0 & \mathcal{H} & \widehat{K}_1 \\ \widehat{T}_{n,1}^T & \widehat{K}_1^T & \widehat{Z}_1 \\ \widehat{T}_{n,2}^T & \widehat{K}_2^T & \widehat{Z}_2^T \end{pmatrix}. \end{aligned}$$

Here, the notation $(*)^T$ indicates that the corresponding equality follows from transposing the entire matrix. To prove (3.33), (3.36) implies that it suffices to show that

$$(3.37) \quad \text{rank} \begin{pmatrix} \mathcal{H}_0 & \mathcal{H} & \widehat{K}_1 \\ \widehat{T}_{n,1}^T & \widehat{K}_1^T & \widehat{Z}_1 \\ \widehat{T}_{n,2}^T & \widehat{K}_2^T & \widehat{Z}_2^T \end{pmatrix} = \text{rank } M_{\mathcal{T}}(n).$$

By (3.20), (3.35) and the ordering of the columns (see (3.3)) of the matrix

$$\begin{pmatrix} \mathcal{H}_0^T & \widehat{T}_{n,1} & \widehat{T}_{n,2} \\ \mathcal{H} & \widehat{K}_1 & \widehat{K}_2 \end{pmatrix},$$

note that

$$(3.38) \quad \text{rank } M_{\mathcal{T}}(n) = \begin{pmatrix} \mathcal{H}_0^T & \widehat{T}_{n,1} & \widehat{T}_{n,2} \\ \mathcal{H} & \underbrace{\widehat{K}_1 \quad \widehat{K}_2}_{p_0 \text{ columns}} \end{pmatrix} = \text{rank} \begin{pmatrix} \mathcal{H}_0^T \\ \mathcal{H} \end{pmatrix} + p_0.$$

We permute the columns of $\begin{pmatrix} \mathcal{H}_0^T \\ \mathcal{H} \end{pmatrix}$ to $\begin{pmatrix} \mathcal{H}_0^T \\ \mathcal{H} \end{pmatrix} P_2 =: \begin{pmatrix} \mathcal{H}_{0,1}^T & \mathcal{H}_{0,2}^T \\ \mathcal{H}_1^T & \mathcal{H}_2^T \end{pmatrix}$, using a permutation matrix $P_2 \in M_{np}(\mathbb{R})$, such that

$$(3.39) \quad \text{rank} \begin{pmatrix} \mathcal{H}_{0,1} & \mathcal{H}_1 \\ \mathcal{H}_{0,2} & \mathcal{H}_2 \end{pmatrix} \underbrace{=}_{(*)^T} \text{rank} \begin{pmatrix} \mathcal{H}_0^T \\ \mathcal{H}^T \end{pmatrix} = \text{rank} \begin{pmatrix} \mathcal{H}_{0,2}^T \\ \mathcal{H}_2^T \end{pmatrix} \underbrace{=}_{(*)^T} \text{rank} \begin{pmatrix} \mathcal{H}_{0,2} & \mathcal{H}_2 \end{pmatrix}$$

and

$$(3.40) \quad \text{Ker} \begin{pmatrix} \mathcal{H}_{0,2}^T & \widehat{T}_{n,1} & \widehat{T}_{n,2} \\ \mathcal{H}_2^T & \widehat{K}_1 & \widehat{K}_2 \end{pmatrix} \underbrace{=}_{(3.38)} \{0\}.$$

Apply the same permutation P_2 on $\widehat{K}_1^T$ to obtain $\widehat{K}_1^T P_2 =: ((\widehat{K}_1^{(1)})^T \quad (\widehat{K}_1^{(2)})^T)$. It follows from (3.40) that

$$\mathcal{C} \begin{pmatrix} \mathcal{H}_{0,2} & \mathcal{H}_2 \\ \widehat{T}_{n,1}^T & \widehat{K}_1^T \\ \widehat{T}_{n,2}^T & \widehat{K}_2^T \end{pmatrix} = \mathbb{R}^{\text{rank } M_{\mathcal{T}}(n)} \quad \text{and} \quad \begin{pmatrix} \mathcal{H}_{0,2} & \mathcal{H}_2 \\ \widehat{T}_{n,1}^T & \widehat{K}_1^T \\ \widehat{T}_{n,2}^T & \widehat{K}_2^T \end{pmatrix} \text{ is surjective,}$$

whence

$$(3.41) \quad \text{rank} \begin{pmatrix} \mathcal{H}_{0,2} & \mathcal{H}_2 & \widehat{K}_1^{(2)} \\ \widehat{T}_{n,1}^T & \widehat{K}_1^T & \widehat{Z}_1 \\ \widehat{T}_{n,2}^T & \widehat{K}_2^T & \widehat{Z}_2 \end{pmatrix} = \text{rank} \begin{pmatrix} \mathcal{H}_{0,2} & \mathcal{H}_2 \\ \widehat{T}_{n,1}^T & \widehat{K}_1^T \\ \widehat{T}_{n,2}^T & \widehat{K}_2^T \end{pmatrix}.$$

By Theorem 2.3, choose a finitely atomic representing measure ν for $\mathcal{T}$ and define

$$Z_\ell^{(e)} := \int_{\mathbb{R}} x^{2n+\ell} d\nu, \quad \ell = 1, 2.$$

Then the moment matrix corresponding to $(T_0, \dots, T_{2n}, Z_1^{(e)}, Z_2^{(e)})$ is positive semidefinite. Using the notation from (3.16), this matrix can be written as

$$\begin{pmatrix} \mathcal{H}_0 & \mathcal{H} & K \\ T_n & K^T & Z_1^{(e)} \\ T_{n+1} & \text{row}(\text{row}(T_i)_{i \in [n+2; 2n]}, Z_1^{(e)}) & Z_2^{(e)} \end{pmatrix}.$$

By Theorem 2.13, used for

$$A := \begin{pmatrix} \mathcal{H}_0 & \mathcal{H} \\ T_n & K^T \end{pmatrix}, \quad B := \begin{pmatrix} K \\ Z_1^{(e)} \end{pmatrix} = \text{col}(\text{col}(T_i)_{i \in [n+1; 2n]}, Z_1^{(e)}), \quad C := Z_2^{(e)},$$

it follows that $\mathcal{C}(B) \subseteq \mathcal{C}(A)$. In particular, $\text{rank}(\mathcal{H}_0 \quad \mathcal{H}) = \text{rank}(\mathcal{H}_0 \quad \mathcal{H} \quad K)$, whence

$$(3.42) \quad \text{rank} \begin{pmatrix} \mathcal{H}_{0,1} & \mathcal{H}_1 \\ \mathcal{H}_{0,2} & \mathcal{H}_2 \end{pmatrix} = \text{rank} \begin{pmatrix} \mathcal{H}_{0,1} & \mathcal{H}_1 & \widehat{K}_1^{(1)} \\ \mathcal{H}_{0,2} & \mathcal{H}_2 & \widehat{K}_1^{(2)} \end{pmatrix}.$$

By Lemma 2.15, used for

$$\begin{aligned} A_1 &:= (\mathcal{H}_{0,1} \quad \mathcal{H}_1), & B_1 &:= \widehat{K}_1^{(1)}, \\ A_2 &:= (\mathcal{H}_{0,2} \quad \mathcal{H}_2), & B_2 &:= \widehat{K}_1^{(2)}, \\ C &:= \begin{pmatrix} \widehat{T}_{n,1}^T & \widehat{K}_1^T \\ \widehat{T}_{n,2}^T & \widehat{K}_2^T \end{pmatrix}, & D &:= \begin{pmatrix} \widehat{Z}_1 \\ \widehat{Z}_2^T \end{pmatrix}, \end{aligned}$$

and the rank equalities (3.39), (3.41) and (3.42), we get

$$(3.43) \quad \text{rank} \begin{pmatrix} \mathcal{H}_{0,1} & \mathcal{H}_1 & \widehat{K}_1^{(1)} \\ \mathcal{H}_{0,2} & \mathcal{H}_2 & \widehat{K}_1^{(2)} \\ \widehat{T}_{n,1}^T & \widehat{K}_1^T & \widehat{Z}_1 \\ \widehat{T}_{n,2}^T & \widehat{K}_2^T & \widehat{Z}_2^T \end{pmatrix} = \text{rank} \begin{pmatrix} \mathcal{H}_{0,1} & \mathcal{H}_1 \\ \mathcal{H}_{0,2} & \mathcal{H}_2 \\ \widehat{T}_{n,1}^T & \widehat{K}_1^T \\ \widehat{T}_{n,2}^T & \widehat{K}_2^T \end{pmatrix} \underbrace{=}_{(3.38)} \text{rank } M_{\mathcal{T}}(n).$$

Hence, the equality (3.37) holds, which concludes the proof of Claim 5. ■

By Claim 5,

$$(3.44) \quad \begin{pmatrix} \widehat{K}_1 \\ \widehat{Z}_1 \\ \widehat{Z}_2^T \\ \widehat{Z}_3^T \end{pmatrix} = \begin{pmatrix} \mathcal{H}_0 & \mathcal{H} \\ \widehat{T}_{n,1}^T & \widehat{K}_1^T \\ \widehat{T}_{n,2}^T & \widehat{K}_2^T \\ \widehat{T}_{n,3}^T & \widehat{K}_3^T \end{pmatrix} \begin{pmatrix} U_1 \\ U_2 \end{pmatrix}$$

for some real matrices $U_1 \in M_{p \times (p_0 - m + \text{card } \mathcal{A})}(\mathbb{R})$ and $U_2 \in M_{np \times (p_0 - m + \text{card } \mathcal{A})}(\mathbb{R})$ with

$$(3.45) \quad \text{rank } U_1 = p_0 - m + \text{card } \mathcal{A} \quad (\text{see (3.29)}),$$

or equivalently

$$(3.46) \quad \begin{pmatrix} \mathcal{H}_0 & \mathcal{H} & \widehat{K}_1 \\ \widehat{T}_{n,1}^T & \widehat{K}_1^T & \widehat{Z}_1 \\ \widehat{T}_{n,2}^T & \widehat{K}_2^T & \widehat{Z}_2^T \\ \widehat{T}_{n,3}^T & \widehat{K}_3^T & \widehat{Z}_3^T \end{pmatrix} \begin{pmatrix} -U_1 \\ -U_2 \\ I_{p_0 - m + \text{card } \mathcal{A}} \end{pmatrix} = \mathbf{0}_{(n+1)p \times (p_0 - m + \text{card } \mathcal{A})}.$$

By (3.26) and (3.46), we have that

$$\begin{pmatrix} \mathcal{H}_0 & \mathcal{H} & \widehat{K} \\ \widehat{T}_n^T & \widehat{K}^T & \widehat{Z} \end{pmatrix} \begin{pmatrix} -U_1 & \mathbf{0}_{p \times (m - \text{card } \mathcal{A})} & \mathbf{0}_{p \times (p - p_0)} \\ -U_2 & -J_1 & -\widehat{G}_1 \\ I_{p_0 - m + \text{card } \mathcal{A}} & -J_2 & -\widehat{G}_2 \\ \mathbf{0} & I_{m - \text{card } \mathcal{A}} & -\widehat{G}_3 \\ \mathbf{0} & \mathbf{0} & I_{p - p_0} \end{pmatrix} = \mathbf{0}_{(n+1)p \times p}.$$

Therefore

$$(3.47) \quad \begin{pmatrix} \widehat{K} \\ \widehat{Z} \end{pmatrix} \widehat{Q}_n = \text{row}(\widehat{X}_{\mathcal{T}}^i)_{i \in [n]} \begin{pmatrix} U_2 & J_1 & \widehat{G}_1 \end{pmatrix} + \widehat{X}_{\mathcal{T}}^0 \begin{pmatrix} U_1 & \mathbf{0}_{p \times (m - \text{card } \mathcal{A} + p - p_0)} \end{pmatrix},$$

where

$$(3.48) \quad \widehat{Q}_n := \begin{pmatrix} I_{p_0 - m + \text{card } \mathcal{A}} & -J_2 & -\widehat{G}_2 \\ \mathbf{0} & I_{m - \text{card } \mathcal{A}} & -\widehat{G}_3 \\ \mathbf{0} & \mathbf{0} & I_{p - p_0} \end{pmatrix}$$

and

$$(3.49) \quad \text{row}(\hat{X}_{\mathcal{T}}^i)_{i \in [0;n]} := (I_{np} \oplus P^T) \text{row}(X_{\mathcal{T}}^i)_{i \in [0;n]} = \begin{pmatrix} \mathcal{H}_0 & \mathcal{H} \\ \hat{T}_n^T & \hat{K}^T \end{pmatrix}.$$

Note that $\hat{G}_1 = \text{col}(H_i \check{H}_n^{-1})_{i \in [0;n-1]}$ and $\text{col}(\hat{G}_2, \hat{G}_3) = P_1^T \check{H}_{n,(0)} \check{H}_n^{-1}$. Writing

$$(3.50) \quad \begin{pmatrix} U_2 & J_1 \end{pmatrix} =: \begin{pmatrix} U_{2,0} & J_{1,0} \\ U_{2,1} & J_{1,1} \\ \vdots & \vdots \\ U_{2,n-1} & J_{1,n-1} \end{pmatrix},$$

where $U_{2,i} \in M_{p \times (p_0 - m + \text{card } \mathcal{A})}(\mathbb{R})$ and $J_{1,i} \in M_{p \times (m - \text{card } \mathcal{A})}(\mathbb{R})$, and defining

$$(3.51) \quad Q_i := \begin{pmatrix} U_{2,i} & J_{1,i} & H_i \check{H}_n^{-1} \end{pmatrix} \quad \text{for } i \in [0; n-1],$$

(3.47) is equivalent to

$$(3.52) \quad \begin{pmatrix} \hat{K} \\ \hat{Z} \end{pmatrix} \hat{Q}_n = \sum_{i=1}^n \hat{X}_{\mathcal{T}}^i Q_{i-1} + \hat{X}_{\mathcal{T}}^0 (U_1 \quad \mathbf{0}_{p \times (m - \text{card } \mathcal{A} + p - p_0)}).$$

Using (3.49) and

$$\begin{pmatrix} \hat{K} \\ \hat{Z} \end{pmatrix} = \begin{pmatrix} KP \\ P^T Z P \end{pmatrix} = (I_{np} \oplus P^T) \begin{pmatrix} \text{col}(T_i)_{i \in [n+1; 2n]} \\ Z \end{pmatrix} P$$

in (3.52), we get

$$(3.53) \quad \begin{pmatrix} \text{col}(T_i)_{i \in [n+1; 2n]} \\ Z \end{pmatrix} Q_n = \sum_{i=1}^n X_{\mathcal{T}}^i Q_{i-1} + X_{\mathcal{T}}^0 (U_1 \quad \mathbf{0}_{p \times (m - \text{card } \mathcal{A} + p - p_0)}),$$

where

$$(3.54) \quad Q_n := P \hat{Q}_n.$$

We now define

$$T_{2n+1} := Z \quad \text{and} \quad T_{2n+2} := \text{row}(T_i)_{i \in [n+1; 2n+1]} (M_{\mathcal{T}}(n))^{\dagger} \text{col}(T_i)_{i \in [n+1; 2n+1]}.$$

Let

$$(3.55) \quad \begin{aligned} \mathcal{T}^{(e)} &:= (T_i)_{i \in [0; 2n+2]}, \quad S_{2n+\ell} := L_{\mathcal{T}^{(e)}}((x+t)^{2n+\ell}), \quad \ell = 1, 2, \quad \text{and} \\ \mathcal{S}^{(e)} &:= (S_i)_{i \in [0; 2n+2]}. \end{aligned}$$

Define a matrix polynomial

$$(3.56) \quad H(x) := x^{n+1} Q_n - \sum_{i=1}^n x^i Q_{i-1} - (U_1 \quad \mathbf{0}_{p \times (m - \text{card } \mathcal{A} + p - p_0)}),$$

By (3.53) and the definition of $\mathcal{T}^{(e)}$, we have that $H(x)$ is a block column relation of $M_{\mathcal{T}^{(e)}}(n+1)$. By Corollary 2.9, $H(x-t)$ is a block column relation of $M_{\mathcal{S}^{(e)}}(n+1)$. By (3.45), we have

$$(3.57) \quad \text{rank} \begin{pmatrix} U_1 & \mathbf{0}_{p \times (m - \text{card } \mathcal{A} + p - p_0)} \end{pmatrix} = p_0 - m + \text{card } \mathcal{A}.$$

Let

$$\widetilde{W}_0 := \text{Ker} \begin{pmatrix} U_1 & \mathbf{0}_{p \times (m - \text{card } \mathcal{A} + p - p_0)} \end{pmatrix} \quad \text{and} \quad \widetilde{W}_i := \widetilde{W}_0 \bigcap_{j=0}^{i-1} \text{Ker } Q_j \quad \text{for } i \in [n].$$

Note that, for each $i \in [0; n-1]$, Q_i has $H_i \check{H}_n^{-1}$ as its final block of columns (see (3.51)), and that this block has $p - p_0$ columns. Hence, the following holds:

$$(3.58) \quad \text{Let } i \in [0; n-1]. \text{ If } v \in \bigcap_{j=0}^i \text{Ker}(H_j \check{H}_n^{-1}), \text{ then } \begin{pmatrix} 0 \\ v \end{pmatrix} \in \widetilde{W}_{i+1}.$$

Let $\widetilde{s}_i := \dim \widetilde{W}_i$ for $i \in [0; n]$. Recall the definition of s_i from Claim 2. Since $\check{H}_n$ is invertible, for each $i \in [0; n-1]$,

$$\bigcap_{j=0}^i \text{Ker}(H_j \check{H}_n^{-1}) = \check{H}_n \left(\bigcap_{j=0}^i \text{Ker } H_j \right).$$

Consequently, this intersection has dimension s_i . By (3.58), it follows that $\widetilde{s}_i \geq s_{i-1}$ for $i \in [n]$. Therefore

$$(3.59) \quad \begin{aligned} \sum_{i=0}^n \widetilde{s}_i &\geq \widetilde{s}_0 + \sum_{i=0}^{n-1} s_i \geq m - \text{card } \mathcal{A} + p - p_0 \\ &\quad + (n - z_s)(p - p_0) + \sum_{j=1}^{s-1} (z_{j+1} - z_j)(p - p_0 - r_{s-j}) + \text{card } \mathcal{A} \\ &= m + (n+1)p - ((n+1)p_0 + \sum_{j=1}^s z_{s-j+1}p_j) \\ &= m + (n+1)p - \text{rank } M(n), \end{aligned}$$

where we used (3.57) and (3.9) in the second inequality and Claim 1 in the last equality. By Lemma 2.11, the invertibility of Q_n (see (3.48) and (3.54)) and (3.59), it follows that

$$(3.60) \quad \det H(x - t) = (x - t)^{m+(n+1)p - \text{rank } M(n)} g(x),$$

where $0 \neq g(x) \in \mathbb{R}[x]$ and $\deg(\det H(x - t)) = (n+1)p$ (since Q_n is invertible). Therefore $\deg g(x) = \text{rank } M(n) - m$. By Theorem 2.3, choose a $(\text{rank } M(n))$ -atomic representing measure μ for $\mathcal{S}^{(e)}$. In particular, μ is a minimal representing measure for L . Set $A_{\{t\}} := \mu(\{t\})$ and write

$$\mu = \delta_t A_{\{t\}} + \sum_{j=1}^{\ell} \delta_{x_j} A_j,$$

where $\ell \in \mathbb{N} \cup \{0\}$, the points $x_j \in \mathbb{R} \setminus \{t\}$ are pairwise distinct, and $A_j \in \mathbb{S}_p^{\geq 0}(\mathbb{R})$. Here, $A_{\{t\}}$ is allowed to be zero, and

$$\text{rank } A_{\{t\}} + \sum_{j=1}^{\ell} \text{rank } A_j = \text{rank } M(n).$$

Lemma 2.5 and (3.60) give

$$\text{rank } M(n) - \text{rank } A_{\{t\}} = \sum_{j=1}^{\ell} \text{rank } A_j \leq \sum_{j=1}^{\ell} \text{mult}_g x_j \leq \deg g = \text{rank } M(n) - m.$$

Thus $\text{rank } A_{\{t\}} \geq m$.

Next, we need to establish the following claim.

Claim 6. $\text{rank } A_{\{t\}} = m$.

Proof of Claim 6. Suppose, on the contrary, that $\text{rank } A_{\{t\}} = m'$ for some

$$(3.61) \quad m' > m.$$

Define $\tilde{\mu} := \mu - \delta_t A_{\{t\}}$. Let $\tilde{L} : \mathbb{R}[x] \rightarrow \mathbb{S}_p(\mathbb{R})$ be a linear operator, defined by

$$\tilde{L}(x^i) \equiv \tilde{S}_i := \int_{\mathbb{R}} x^i d\tilde{\mu} \quad \text{for } i \in \mathbb{N} \cup \{0\}.$$

Let $\tilde{\mathcal{S}} := (\tilde{S}_i)_{i \in [0; 2n+2]}$. We have

$$(3.62) \quad \begin{aligned} \text{rank } M(n) - m' &\leq \text{rank } M_{\tilde{\mathcal{S}}}(n) \leq \text{rank } M_{\tilde{\mathcal{S}}}(n+1), \\ \text{rank } M_{\tilde{\mathcal{S}}}(n+1) &\leq \sum_{j=1}^{\ell} \text{rank } A_j = \text{rank } M(n) - m', \end{aligned}$$

where the first inequality follows from the fact that the difference $M(n) - M_{\tilde{\mathcal{S}}}(n)$ is a moment matrix of the linear operator corresponding to the matrix measure $\delta_t A_{\{t\}}$. The inequalities (3.62) imply that $\text{rank } M(n) - m' \leq \text{rank } M_{\tilde{\mathcal{S}}}(n+1) \leq \text{rank } M(n) - m'$, whence all inequalities in (3.62) must be equalities. In particular,

$$(3.63) \quad \text{rank } M_{\tilde{\mathcal{S}}}(n) = \text{rank } M_{\tilde{\mathcal{S}}}(n+1) = \text{rank } M(n) - m'.$$

For $i \in [0; 2n+2]$ we define $\tilde{T}_i := \int_{\mathbb{R}} (x-t)^i d\tilde{\mu}$. Note that

$$(3.64) \quad \tilde{T}_i = \int_{\mathbb{R}} (x-t)^i d(\tilde{\mu} + \delta_t A_{\{t\}}) = T_i \quad \text{for } i \in [2n+2].$$

Define the sequence $\tilde{\mathcal{T}} := (\tilde{T}_i)_{i \in [0; 2n+2]}$ and write

$$X_{\tilde{\mathcal{T}}}^i \equiv \text{row}((x_{\tilde{\mathcal{T}}}^i)_j)_{j \in [p]} := \text{col}(\tilde{T}_{i+j})_{j \in [0; n+1]}$$

for $i \in [0; n+1]$. By (3.63) and Proposition 2.8(5) we have

$$(3.65) \quad \begin{aligned} m' &= \text{rank } M_{\mathcal{S}}(n+1) - \text{rank}(\text{row}(X_{\tilde{\mathcal{T}}}^i)_{i \in [0; n+1]}) \\ &= \text{rank } M_{\mathcal{S}}(n) - \text{rank}(\text{row}(X_{\tilde{\mathcal{T}}}^i)_{i \in [0; n+1]}). \end{aligned}$$

By (3.64) and Claim 1, it follows that

$$(3.66) \quad \text{rank row}(X_{\tilde{\mathcal{T}}}^i)_{i \in [n]} = \text{rank row}(X_{\mathcal{T}}^i)_{i \in [n]} = \text{rank } M_{\mathcal{T}}(n) - p_0 - \text{card } \mathcal{A}.$$

Further,

$$(3.67) \quad \begin{aligned} \text{rank row}(X_{\tilde{\mathcal{T}}}^i)_{i \in [n+1]} &\stackrel{(3.64)}{=} \text{rank row}(X_{\mathcal{T}}^i)_{i \in [n+1]} \\ &\stackrel{\text{Claim 3}}{=} \text{rank row}(X_{\mathcal{T}}^i)_{i \in [n]} + p_0 - m + \text{card } \mathcal{A} \\ &\stackrel{(3.66)}{=} \text{rank } M_{\mathcal{T}}(n) - m. \end{aligned}$$

Finally,

$$\text{rank } M_{\mathcal{T}}(n) - m \stackrel{(3.67)}{=} \text{rank row}(X_{\tilde{\mathcal{T}}}^i)_{i \in [n+1]} \stackrel{(3.65)}{\leq} \text{rank } M_{\mathcal{T}}(n) - m',$$

which implies that $m' \leq m$ and contradicts (3.61). This proves Claim 6. $\blacksquare$

It remains to establish the moreover part. Note that the representing measure is unique if and only if $\hat{Z}$, satisfying (3.25) and (3.26), is unique. Indeed, in this case the polynomial $H(x)$ as in (3.56) is uniquely determined and it completely determines the minimal measure for the extended sequence.

First note that in the decomposition (3.28) there is no freedom in choosing

$$\widehat{Z}_1 \in \mathbb{S}_{p_0-m+\text{card } \mathcal{A}}(\mathbb{R})$$

such that (3.29) holds if and only if $p_0 - m + \text{card } \mathcal{A} = 0$, or equivalently $m = \text{card } \mathcal{A} + p_0$. Since $\mathcal{B} \subseteq ([p] \setminus \text{Dep}(X_{\mathcal{T}}^{z_s}))$, it follows by definition of p_0 (see (3.4)) that this is further equivalent to $\mathcal{B} = [p] \setminus \text{Dep}(X_{\mathcal{T}}^{z_s})$ and $m = \text{card}(\mathcal{A} \cup \mathcal{B})$, which are the first and the second condition in (1.4).

By the previous paragraph, the matrices $\widehat{Z}_1$, $\widehat{Z}_2$ and $\widehat{Z}_3$ in (3.28) are empty. Next, observe the matrices $\widehat{Z}_4$ and $\widehat{Z}_5$ in (3.28). By (3.30) and (3.31), $\widehat{Z}_4$ and $\widehat{Z}_5$ will be unique if and only if $\widehat{K}_2^T v = \widehat{K}_3^T v = \mathbf{0}$ for any $v \in \text{Ker } \mathcal{H}$. This is due to the fact that J in (3.23) is unique up to the addition of a matrix with each column from $\text{Ker } \mathcal{H}$ (note that $\widehat{K}_1$ is an empty matrix). Note that $\text{Ker } \mathcal{H} \subseteq \text{Ker } \widehat{K}_2^T$ and $\text{Ker } \mathcal{H} \subseteq \text{Ker } \widehat{K}_3^T$ are exactly the third and the fourth condition in (1.4).

Finally, we observe the matrix $\widehat{Z}_6$ in (3.28). It remains to prove that the four conditions in (1.4) ensure $\widehat{Z}_6$ is also unique. By (3.30), it suffices to prove that $\begin{pmatrix} \widehat{K}_3^T & \widehat{Z}_5^T \end{pmatrix} \widehat{G}$ does not depend on the choice of $\widehat{G}$ satisfying (3.22). If $\widehat{G}$ and $\widehat{G}'$ both satisfy (3.22), then

$$\mathbf{0} = \begin{pmatrix} \mathcal{H} & \widehat{K}_2 \end{pmatrix} (\widehat{G} - \widehat{G}') \underbrace{=}_{(3.23)} \mathcal{H} \begin{pmatrix} I & J \end{pmatrix} (\widehat{G} - \widehat{G}'),$$

whence

$$(3.68) \quad \text{every column of the matrix } \begin{pmatrix} I & J \end{pmatrix} (\widehat{G} - \widehat{G}') \text{ is in } \text{Ker } \mathcal{H}.$$

Note that

$$\begin{pmatrix} \widehat{K}_3^T & \widehat{Z}_5^T \end{pmatrix} (\widehat{G} - \widehat{G}') \underbrace{=}_{(3.31)} \widehat{K}_3^T \begin{pmatrix} I & J \end{pmatrix} (\widehat{G} - \widehat{G}') \underbrace{=}_{\substack{(3.68), \\ \text{Ker } \mathcal{H} \subseteq \text{Ker } \widehat{K}_3^T}} \mathbf{0},$$

whence $\begin{pmatrix} \widehat{K}_3^T & \widehat{Z}_5^T \end{pmatrix} \widehat{G}$ does not depend on the choice of $\widehat{G}$ satisfying (3.22). This proves the moreover part of Theorem 1.1.

4. PROOF OF THE IMPLICATION (1) $\Rightarrow$ (2) OF THEOREM 1.1

Let $\mu = \delta_t A_1 + \sum_{j=2}^{\ell} \delta_{x_j} A_j$ be a finitely atomic $p \times p$ matrix representing measure for L with $\ell \in \mathbb{N}$, $x_j \in \mathbb{R} \setminus \{t\}$, $A_j \in \mathbb{S}_p^{\geq 0}(\mathbb{R})$, $\sum_{j=1}^{\ell} \text{rank } A_j = \text{rank } M(n)$ and $\text{mult}_t \mu = m$. For $k = 1, 2$ define $S_{2n+k} := \int_{\mathbb{R}} x^{2n+k} d\mu$ and $T_{2n+k} := \int_{\mathbb{R}} (x-t)^{2n+k} d\mu$. Let $\tilde{\mu} := \sum_{j=2}^{\ell} \delta_{x_j} A_j$ and $\tilde{L} : \mathbb{R}[x] \rightarrow \mathbb{S}_p(\mathbb{R})$ be a linear operator, defined by

$$\tilde{L}(x^i) \equiv \tilde{S}_i := \int_{\mathbb{R}} x^i d\tilde{\mu} \quad \text{for } i \in \mathbb{N} \cup \{0\}.$$

Write $\tilde{\mathcal{S}} := (\tilde{S}_i)_{i \in [0; 2n+2]}$.

Analogously to the proof of Claim 6 in the proof of the implication (2) $\Rightarrow$ (1) of Theorem 1.1, it follows that

$$(4.1) \quad \text{rank } M_{\tilde{\mathcal{S}}}(n) = \text{rank } M_{\tilde{\mathcal{S}}}(n+1) = \text{rank } M(n) - m.$$

For $i \in [0; 2n+2]$ we define $\tilde{T}_i := \int_{\mathbb{R}} (x-t)^i d\tilde{\mu}$. Note that

$$\tilde{T}_i = \int_{\mathbb{R}} (x-t)^i d(\tilde{\mu} + \delta_t A_1) = T_i \quad \text{for } i \in [2n+2].$$

Define the sequence $\tilde{\mathcal{T}} := (\tilde{T}_i)_{i \in [0; 2n+2]}$ and for $i \in [0; n+1]$ write

$$X_{\mathcal{T}}^i \equiv \text{row} \left((x_{\mathcal{T}}^{(i)})_j \right)_{j \in [p]} := \text{col}(T_{i+j})_{j \in [0; n+1]},$$

$$X_{\tilde{\mathcal{T}}}^i \equiv \text{row} \left((x_{\tilde{\mathcal{T}}}^{(i)})_j \right)_{j \in [p]} := \text{col}(\tilde{T}_{i+j})_{j \in [0; n+1]}.$$

By (4.1) and Proposition 2.8.(5), we have

$$\begin{aligned} m &= \text{rank } M(n) - \text{rank} \left(\text{row}(X_{\tilde{\mathcal{T}}}^i)_{i \in [0; n+1]} \right) \\ &= \text{rank } M(n) - \text{rank} \left(\text{row}(X_{\tilde{\mathcal{T}}}^i)_{i \in [0; n]} \right). \end{aligned}$$

Let $\text{Dep}(X_*^i)$, $\text{Dep}_1(X_*^i)$ and $\text{DepN}(X_*^i)$, $\text{DepN}_\ell(X_*^i)$, $\ell = 0, 1$, be as in (2.5), (2.6) and (2.7), respectively. Define

$$\mathcal{D} := \{i \in [0; n+1] : \text{DepN}(X_{\tilde{\mathcal{T}}}^i) \neq \emptyset\}.$$

Note that $\text{Dep}(X_{\tilde{\mathcal{T}}}^{n+1}) = [p]$, and thus Corollary 2.7 implies that $\mathcal{D} \neq \emptyset$. Hence, writing

$$\mathcal{D} = \{z_1, \dots, z_s : 0 \leq z_1 < \dots < z_s \leq n+1\},$$

the same corollary yields $\text{Dep}(X_{\tilde{\mathcal{T}}}^{z_s}) = [p]$.

Let

$$\begin{aligned} p_{j,\ell} &:= \text{card } \text{DepN}_\ell(X_{\tilde{\mathcal{T}}}^{z_s-j+1}) \quad \text{for } j \in [s] \text{ and } \ell = 0, 1, \\ p_j &:= \text{card } \text{DepN}(X_{\tilde{\mathcal{T}}}^{z_s-j+1}) = p_{j,0} + p_{j,1} \quad \text{for } j \in [s], \end{aligned}$$

where the last equality follows from Lemma 2.2, and

$$\tilde{\mathcal{A}} := \bigcup_{i=1}^{n+1} \text{DepN}_1(X_{\tilde{\mathcal{T}}}^i).$$

Claim 1. We have

$$\text{rank } M_{\tilde{\mathcal{T}}}(n+1) = \text{rank } M_{\tilde{\mathcal{T}}}(n) = \sum_{j=1}^s z_{s-j+1} p_j.$$

Proof of Claim 1. By (4.1) and Proposition 2.8.(5),

$$\text{rank } M_{\tilde{\mathcal{T}}}(n+1) = \text{rank } M_{\tilde{\mathcal{T}}}(n).$$

By Corollary 2.7 and the fact that $\text{Dep}(X_{\tilde{\mathcal{T}}}^{z_s}) = [p]$, we have

$$[p] = \bigsqcup_{j=1}^s \text{DepN}(X_{\tilde{\mathcal{T}}}^{z_s-j+1}).$$

We compute the rank by reading the columns from left to right. For $r \in \text{DepN}(X_{\tilde{\mathcal{T}}}^{z_s-j+1})$, the definition of DepN and the nesting of the dependency sets show that $r \notin \text{Dep}(X_{\tilde{\mathcal{T}}}^i)$ precisely for $i \in [0; z_{s-j+1} - 1]$. Hence each such index contributes exactly z_{s-j+1} columns that are independent of their predecessors. Since this group has cardinality p_j , summing over the partition above yields

$$\text{rank } M_{\tilde{\mathcal{T}}}(n+1) = \sum_{j=1}^s z_{s-j+1} p_j.$$

Combining this formula with $\text{rank } M_{\tilde{\mathcal{T}}}(n+1) = \text{rank } M_{\tilde{\mathcal{T}}}(n)$ completes the proof of Claim 1. ■

Claim 2. $\tilde{\mathcal{A}} = \emptyset$.

Proof of Claim 2. Analogously to the proof of Claim 2 of the implication (2) $\Rightarrow$ (1) of Theorem 1.1 with $\tilde{\mathcal{T}}$ in place of $\mathcal{T}$ and noting that $p_0 = 0$, there exists a matrix polynomial

$$H(x) = x^{n+1} H_{n+1} - \sum_{i=0}^n x^i H_i,$$

such that:

- (1) $H(x)$ is a block column relation of $M_{\tilde{\mathcal{T}}}(n+1)$.
- (2) H_{n+1} is invertible.
- (3) Defining $W_i := \bigcap_{j=0}^i \text{Ker } H_j$ and $s_i := \dim W_i$ for $i \in [0; n]$, we have

$$\begin{aligned}
 \sum_{i=0}^n s_i &\geq (n+1 - z_s)p + \sum_{j=1}^{s-1} (z_{j+1} - z_j)(p - \sum_{k=1}^{s-j} p_k) + \text{card } \tilde{\mathcal{A}} \\
 (4.2) \quad &= (n+1)p - \sum_{j=1}^s z_{s-j+1}p_j + \text{card } \tilde{\mathcal{A}} \\
 &\stackrel{\text{Claim 1}}{=} (n+1)p - \text{rank } M_{\tilde{\mathcal{T}}}(n) + \text{card } \tilde{\mathcal{A}}.
 \end{aligned}$$

By Lemma 2.11, the invertibility of H_{n+1} and (4.2), it follows that

$$\det H(x) = x^{(n+1)p - \text{rank } M_{\tilde{\mathcal{T}}}(n) + \text{card } \tilde{\mathcal{A}}} g(x),$$

where $0 \neq g(x) \in \mathbb{R}[x]$ and $\deg(\det H(x)) = (n+1)p$ (since H_{n+1} is invertible). Therefore $\deg g(x) = \text{rank } M_{\tilde{\mathcal{T}}}(n) - \text{card } \tilde{\mathcal{A}}$. Since $\sum_{j=2}^{\ell} \delta_{x_j-t} A_j$ is a representing measure for $\tilde{\mathcal{T}}$, $\sum_{j=2}^{\ell} \text{rank } A_j = \text{rank } M_{\tilde{\mathcal{T}}}(n)$ and $x_j - t \neq 0$ for each $j \in [2; \ell]$, it follows by Lemma 2.5 that $x_j - t$ are precisely the zeroes of $g(x)$ and $\text{rank } A_j$ are their multiplicities. Therefore $\deg g(x) = \text{rank } M_{\tilde{\mathcal{T}}}(n)$, whence $\text{card } \tilde{\mathcal{A}} = 0$. $\blacksquare$

Let $\mathcal{B}$ be as in Theorem 1.1. Note that

$$(4.3) \quad \text{DepN}_1(X_{\mathcal{T}}^{n+1}) \subseteq \mathcal{B}.$$

Claim 3. For $i \in [0; n]$ we have

$$(4.4) \quad \text{DepN}_1(X_{\mathcal{T}}^{i+1}) \sqcup \text{Dep}(X_{\mathcal{T}}^i) = \text{Dep}(X_{\tilde{\mathcal{T}}}^i).$$

Proof of Claim 3. First we prove the inclusion ($\subseteq$) in (4.4). Fix $i \in [0; n]$. Since $M_{\mathcal{T}}(n+1) \succeq M_{\tilde{\mathcal{T}}}(n+1)$, Lemma 2.12 (used for $A = M_{\mathcal{T}}(n+1)$, $B = M_{\tilde{\mathcal{T}}}(n+1)$) implies that

$$(4.5) \quad \text{DepN}_1(X_{\mathcal{T}}^{i+1}) \subseteq \text{Dep}(X_{\tilde{\mathcal{T}}}^{i+1}) \quad \text{and} \quad \text{Dep}(X_{\mathcal{T}}^i) \subseteq \text{Dep}(X_{\tilde{\mathcal{T}}}^i).$$

Using Claim 2 and Lemma 2.2,

$$\text{DepN}_1(X_{\mathcal{T}}^{i+1}) \subseteq \text{Dep}_1(X_{\tilde{\mathcal{T}}}^{i+1}) \setminus \text{DepN}_1(X_{\tilde{\mathcal{T}}}^{i+1}),$$

whence

$$(4.6) \quad \text{DepN}_1(X_{\mathcal{T}}^{i+1}) \subseteq \text{Dep}(X_{\tilde{\mathcal{T}}}^i).$$

By (4.6) and (4.5), the inclusion ($\subseteq$) in (4.4) follows.

It remains to prove the inclusion ($\supseteq$) in (4.4). Fix $i \in [0; n]$. Let $j \in \text{Dep}(X_{\tilde{\mathcal{T}}}^i)$. We separate two cases.

Case 1: $j \in \text{DepN}_1(X_{\mathcal{T}}^i)$. Since $\text{DepN}_1(X_{\mathcal{T}}^i) \subseteq \text{Dep}(X_{\mathcal{T}}^i)$ by Lemma 2.2, it follows that $j \in \text{Dep}(X_{\mathcal{T}}^i)$.

Case 2: $j \notin \text{DepN}_1(X_{\mathcal{T}}^i)$. Assume also that $j \notin \text{Dep}(X_{\mathcal{T}}^i)$. Since $M_{\tilde{\mathcal{T}}}(n+1)$ is block-recursively generated, it follows that $j \in \text{Dep}_1(X_{\tilde{\mathcal{T}}}^{i+1})$ and hence the equalities $X_{\tilde{\mathcal{T}}}^k = X_{\tilde{\mathcal{T}}}^j$ for $k \in [n+1]$ imply $j \in \text{DepN}_1(X_{\tilde{\mathcal{T}}}^{i+1})$.

This proves the inclusion $(\supseteq)$ in (4.4). ■

We have

$$\begin{aligned}
m &= \text{rank } M_{\mathcal{T}}(n) - \text{rank } M_{\tilde{\mathcal{T}}}(n) \\
&= \left((n+1)p - \sum_{i=0}^n \text{card} \left(\text{Dep}(X_{\mathcal{T}}^i) \right) \right) - \left((n+1)p - \sum_{i=0}^n \text{card} \left(\text{Dep}(X_{\tilde{\mathcal{T}}}^i) \right) \right) \\
&\stackrel{\text{Claim 3}}{=} \sum_{i=0}^n \text{card} \left(\text{DepN}_1(X_{\mathcal{T}}^{i+1}) \right) \\
&= \text{card } \mathcal{A} + \text{card} \left(\text{DepN}_1(X_{\mathcal{T}}^{n+1}) \right) \stackrel{(4.3)}{\leq} \text{card}(\mathcal{A} \cup \mathcal{B}).
\end{aligned}$$

Hence, $\text{card } \mathcal{A} \leq m \leq \text{card}(\mathcal{A} \cup \mathcal{B})$, which proves the implication (1) $\Rightarrow$ (2) of Theorem 1.1.

5. PROOF OF THEOREM 1.5

The equivalence (1) $\Leftrightarrow$ (2) follows directly from Theorem 1.1 for $m = 0$.

Let us prove the implication (1) $\Rightarrow$ (3). Let μ be a minimal representing measure for L such that $\text{mult}_{\mu} t = 0$ and let $L((x-t)^i) := \int_{\mathbb{R}} (x-t)^i d\mu$, $i \in \{-2, -1, 2n+1, 2n+2\}$, be the extension of L to $\text{span}\{(x-t)^{-2}, (x-t)^{-1}, 1, x-t, \dots, (x-t)^{2n+1}, (x-t)^{2n+2}\}$. Write $T_i := L((x-t)^i)$ for $i \in [-2; 2n+2]$. Since $(T_i)_{i=-2}^{2n+2}$ has a representing measure, it follows that the matrix $\widetilde{\mathcal{M}}_1 := (T_{i+j-2})_{i,j=0}^{n+2}$ is positive semidefinite.

Below, each $*$ denotes the corresponding block of $\widetilde{\mathcal{M}}_1$, determined by the moments T_i and the displayed block partition. The explicit entries of these blocks are not needed in the argument. Note that

$$\widetilde{\mathcal{M}}_1 = \begin{pmatrix} T_{-2} & * & * \\ * & \mathcal{M}_2 & * \\ * & * & * \end{pmatrix} = \begin{pmatrix} * & * & * \\ * & \mathcal{M}_3 & * \\ * & * & * \end{pmatrix}.$$

(The blocks in the second matrix are grouped by curly braces below them: $\underbrace{\quad}_{p \text{ columns}}$, $\underbrace{\quad}_{np \text{ columns}}$, $\underbrace{\quad}_{2p \text{ columns}}$.)

Let $w \in \text{Ker } \mathcal{M}_2$. By Lemma 2.14 (used for $A = \widetilde{\mathcal{M}}_1 \in \mathbb{S}_{(n+3)p}(\mathbb{R})$ and $A|_Q = \mathcal{M}_2$ where $Q = [p+1; (n+1)p]$), it follows that $\text{col}(\mathbf{0}_{p \times 1}, w, \mathbf{0}_{2p \times 1}) \in \text{Ker } \widetilde{\mathcal{M}}_1$, whence $w \in \text{Ker } \mathcal{M}_3$. Therefore, $\text{Ker } \mathcal{M}_2 \subseteq \text{Ker } \mathcal{M}_3$.

Similarly

$$\widetilde{\mathcal{M}}_1 = \begin{pmatrix} * & * & * \\ * & \mathcal{M}_3 & * \\ * & * & T_{2n+2} \end{pmatrix} = \begin{pmatrix} * & \mathcal{M}_2 & * \\ * & * & * \\ * & * & * \end{pmatrix}.$$

(The blocks in the second matrix are grouped by curly braces below them: $\underbrace{\quad}_{2p \text{ columns}}$, $\underbrace{\quad}_{np \text{ columns}}$, $\underbrace{\quad}_{p \text{ columns}}$.)

Let $w \in \text{Ker } \mathcal{M}_3$. By Lemma 2.14 (used for $A = \widetilde{\mathcal{M}}_1 \in \mathbb{S}_{(n+3)p}(\mathbb{R})$ and $A|_Q = \mathcal{M}_3$ where $Q = [2p+1; (n+2)p]$), it follows that $\text{col}(\mathbf{0}_{2p \times 1}, w, \mathbf{0}_{p \times 1}) \in \text{Ker } \widetilde{\mathcal{M}}_1$, whence $w \in \text{Ker } \mathcal{M}_2$. Therefore, $\text{Ker } \mathcal{M}_3 \subseteq \text{Ker } \mathcal{M}_2$.

It remains to prove the implication (3) $\Rightarrow$ (2). We have to prove that $\mathcal{A} = \emptyset$. Assume, on the contrary, that there exists $i \in \mathcal{A}$. By definition of $\mathcal{A}$, there is $k \in [n]$ such that $\mathbf{y}_{kp+i} = \sum_{\ell=p+1}^{kp+i-1} \gamma_{\ell} \mathbf{y}_{\ell}$ for some $\gamma_{\ell} \in \mathbb{R}$ and in particular, $\mathbf{y}_{(k-1)p+i} \neq \sum_{\ell=1}^{(k-1)p+i-1} \gamma_{\ell+p} \mathbf{y}_{\ell}$. But this implies

$$w = (-\gamma_{p+1}, \dots, -\gamma_{kp+i-1}, 1, 0, \dots, 0) \in \mathbb{R}^{np} \in \text{Ker } \mathcal{M}_3 \setminus \text{Ker } \mathcal{M}_2,$$

which contradicts $\text{Ker } \mathcal{M}_2 = \text{Ker } \mathcal{M}_3$. Hence, $\mathcal{A} = \emptyset$.

6. PROOF OF COROLLARY 1.6

The equivalence (2) $\Leftrightarrow$ (3) is clear. The implication (1) $\Leftarrow$ (2) is trivial, while the implication (1) $\Rightarrow$ (2) follows by [MS24a, Theorem 5.1]. The equivalence (4) $\Leftrightarrow$ (5) follows by noticing that for $\ell = 0, 2$ we have

$$\sum_{i,j=-n_1}^{n_2-1} v_{j+n_1}^T S_{i+j+\ell} v_{i+n_1} = (\text{col}(v_{i+n_1})_{i \in [-n_1; n_2-1]})^T (S_{i+j+\ell})_{i,j=-n_1}^{n_2-1} \text{col}(v_{i+n_1})_{i \in [-n_1; n_2-1]}.$$

Since $(S_{i+j})_{i,j=-n_1}^{n_2-1}$ is positive semidefinite, for $\ell = 0, 2$ we have

$$\sum_{i,j=-n_1}^{n_2-1} v_{j+n_1}^T S_{i+j+\ell} v_{i+n_1} = 0 \quad \Leftrightarrow \quad \text{col}(v_{i+n_1})_{i \in [-n_1; n_2-1]} \in \text{Ker}(S_{i+j+\ell})_{i,j=-n_1}^{n_2-1}.$$

It remains to prove the equivalence (3) $\Leftrightarrow$ (4). Define a sequence $\tilde{S} = (\tilde{S}_0, \tilde{S}_1, \dots, \tilde{S}_{2n_1+2n_2})$, where $\tilde{S}_i := S_{i-2n_1}$.

First we prove the implication (3) $\Rightarrow$ (4). Let $\mu := \sum_{j=1}^{\ell} \delta_{x_j} A_j$ be a minimal representing measure such that $S_i = \int_{\mathbb{R}} x^i d\mu = \sum_{j=1}^{\ell} x_j^i A_j$ for each i . Clearly, $x_j \neq 0$ for each j and $(S_{i+j})_{i,j=-n_1}^{n_2}$ is positive semidefinite. Defining $\tilde{\mu} := \sum_{j=1}^{\ell} \delta_{x_j} (x_j^{-2n_1} A_j)$, we have

$$\tilde{S}_i = S_{i-2n_1} = \sum_{j=1}^{\ell} x_j^{i-2n_1} A_j = \sum_{j=1}^{\ell} x_j^i (x_j^{-2n_1} A_j) = \int_{\mathbb{R}} x^i d\tilde{\mu}.$$

By Theorem 1.5, used for $t = 0$, $\text{Ker}(S_{i+j})_{i,j=-n_1}^{n_2-1} = \text{Ker}(S_{i+j})_{i,j=-n_1+1}^{n_2}$. This proves (3) $\Rightarrow$ (4).

It remains to prove the implication (4) $\Rightarrow$ (3). By assumption, $M_{\tilde{S}}$ is positive semidefinite. The assumption $\text{Ker}(S_{i+j})_{i,j=-n_1}^{n_2-1} = \text{Ker}(S_{i+j})_{i,j=-n_1+1}^{n_2}$ implies that $\mathcal{C}(S_{i+j})_{i,j=-n_1}^{n_2-1} = \mathcal{C}(S_{i+j})_{i,j=-n_1+1}^{n_2}$. In particular, $\mathcal{C}(\text{col}(\tilde{S}_{n_1+n_2+i})_{i \in [n_1+n_2]}) \subseteq \mathcal{C}(M_{\tilde{S}}(n_1+n_2-1))$. By Theorem 2.3, $\tilde{S}$ admits a representing measure. By Theorem 1.5, $\tilde{S}$ has a minimal representing measure $\tilde{\mu} = \sum_{j=1}^{\ell} \delta_{x_j} A_j$ for some $x_j \in \mathbb{R} \setminus \{0\}$ and $A_j \in \mathbb{S}_p^{\geq 0}(\mathbb{R})$. Namely, $\tilde{S}_i = \sum_{j=1}^{\ell} x_j^i A_j$ for each $i \in [0; 2n_1 + 2n_2]$. But then

$$S_i = \tilde{S}_{i+2n_1} = \sum_{j=1}^{\ell} x_j^{i+2n_1} A_j = \sum_{j=1}^{\ell} x_j^i x_j^{2n_1} A_j,$$

whence $\mu := \sum_{j=1}^{\ell} \delta_{x_j} (x_j^{2n_1} A_j)$ is a minimal representing measure in (3).

7. EXAMPLE FOR THEOREM 1.1

The next example¹ shows how to apply Theorem 1.1 to construct a minimal representing measure μ for a linear operator $L : \mathbb{R}[x]_{\leq 2n} \rightarrow \mathbb{S}_p(\mathbb{R})$ such that $\text{mult}_{\mu} t > 0$, where $t \in \mathbb{R}$ is a prescribed number.

Let $p = 4$, $n = 2$ and $\mathcal{S} := (S_i)_{i \in [0;4]}$, where

$$S_0 = \begin{pmatrix} 4 & 0 & 0 & 0 \\ 0 & 7 & 3 & 3 \\ 0 & 3 & 4 & 3 \\ 0 & 3 & 3 & 3 \end{pmatrix}, \quad S_1 = S_3 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & -1 & -1 & -1 \\ 0 & -1 & 0 & -1 \\ 0 & -1 & -1 & -1 \end{pmatrix}, \quad S_2 = S_4 = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 3 & 1 & 1 \\ 0 & 1 & 2 & 1 \\ 0 & 1 & 1 & 1 \end{pmatrix}.$$

¹The *Mathematica* [Wol] file with numerical computations can be found on the link <https://github.com/ZobovicIgor/Matricial-Gaussian-Quadrature-Rules/tree/main>.

Let $L : \mathbb{R}[x]_{\leq 4} \rightarrow \mathbb{S}_4(\mathbb{R})$ be a linear operator, defined by $L(x^i) = S_i$ for each $i \in [0; 4]$. Since $M(2)$ is positive semidefinite and $\mathcal{C}(\text{col}(S_3, S_4)) \subseteq \mathcal{C}(M(1))$, by Theorem 2.3, L has a representing measure. It is easy to check that $\tilde{\mu} := \sum_{j=1}^3 \delta_{\tilde{x}_j} \tilde{A}_j$ is a minimal representing measure for L , where

$$\tilde{x}_1 = -1, \quad \tilde{x}_2 = 0, \quad \tilde{x}_3 = 1, \quad \tilde{A}_1 = (1) \oplus \begin{pmatrix} 2 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}, \quad \tilde{A}_2 = 2\tilde{A}_1, \quad \tilde{A}_3 = I_3 \oplus (0).$$

Let $t = 1$ and T_i be defined as in Theorem 1.1 for $i \in [0; 4]$. A computation reveals that

$$T_0 = (4) \oplus \begin{pmatrix} 7 & 3 & 3 \\ 3 & 4 & 3 \\ 3 & 3 & 3 \end{pmatrix}, \quad T_1 = -4W, \quad T_2 = 6W, \quad T_3 = -10W, \quad T_4 = 18W,$$

where

$$W = (1) \oplus \begin{pmatrix} 2 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}.$$

Define $\mathcal{T} := (T_i)_{i \in [0; 4]}$. Let $\text{Dep}(X_{\mathcal{T}}^i)$, $\text{Dep}_1(X_{\mathcal{T}}^i)$, $\text{DepN}(X_{\mathcal{T}}^i)$ and $\text{DepN}_{\ell}(X_{\mathcal{T}}^i)$, $\ell = 0, 1$, and $\mathcal{A}, \mathcal{B}$ be as in (2.5), (2.6), (2.7) and Theorem 1.1, respectively. We have

$$\begin{aligned} \text{Dep}(X_{\mathcal{T}}^0) &= \emptyset, & \text{Dep}(X_{\mathcal{T}}^1) &= \{4\}, & \text{Dep}(X_{\mathcal{T}}^2) &= \{3, 4\}, \\ \text{Dep}_1(X_{\mathcal{T}}^0) &= \emptyset, & \text{Dep}_1(X_{\mathcal{T}}^1) &= \{4\}, & \text{Dep}_1(X_{\mathcal{T}}^2) &= \{4\}, \\ \text{DepN}(X_{\mathcal{T}}^0) &= \emptyset, & \text{DepN}(X_{\mathcal{T}}^1) &= \{4\}, & \text{DepN}(X_{\mathcal{T}}^2) &= \{3\}, \\ \text{DepN}_1(X_{\mathcal{T}}^0) &= \emptyset, & \text{DepN}_1(X_{\mathcal{T}}^1) &= \{4\}, & \text{DepN}_1(X_{\mathcal{T}}^2) &= \emptyset, \\ \text{DepN}_0(X_{\mathcal{T}}^0) &= \emptyset, & \text{DepN}_0(X_{\mathcal{T}}^1) &= \emptyset, & \text{DepN}_0(X_{\mathcal{T}}^2) &= \{3\}, \end{aligned}$$

and $\mathcal{A} = \{4\}$, $\mathcal{B} = \{1, 2\}$. Thus $\text{card } \mathcal{A} = 1$ and $\text{card}(\mathcal{A} \cup \mathcal{B}) = 3$. It follows from Theorem 1.1 that for each m with $1 \leq m \leq 3$, there exists a representing measure μ for L such that $\text{mult}_{\mu} t = m$. We will construct such a measure for $m = 2$ (note that $\text{mult}_{\tilde{\mu}} t = \text{rank } \tilde{A}_3 = 3$).

Let $s, z_j, p_{j,\ell}, p_j, p_0$ be as in (3.2), (3.4). By the computations above, we have $s = 2$, $z_1 = 1$, $z_2 = 2$, $p_{1,0} = p_{2,1} = 1$, $p_{1,1} = p_{2,0} = 0$, $p_1 = p_2 = 1$ and $p_0 = 2$. Since the columns of each block column $X_{\mathcal{T}}^i$ are already in the order (3.3), we choose the permutation matrix $Q = I_4$, whence $\check{X}_{\mathcal{T}}^i = X_{\mathcal{T}}^i Q = X_{\mathcal{T}}^i$ for each i . Decompose $\check{X}_{\mathcal{T}}^i$ as in (3.6) for each i . Next we will compute the matrices (3.10) satisfying (3.11) by the formulas:

$$\begin{aligned} \text{col}(\check{B}_0^{(1,1)}, \check{B}_{1,(0)}^{(1,1)}, \check{B}_1^{(1,1)}) &= \begin{pmatrix} \check{X}_{\mathcal{T}}^0 & (\check{X}_{\mathcal{T}}^1)_{(0)} & (\check{X}_{\mathcal{T}}^1)_{(1,0)} & (\check{X}_{\mathcal{T}}^1)_{(1,1)} & (\check{X}_{\mathcal{T}}^1)_{(2,0)} \end{pmatrix}^{\dagger} (\check{X}_{\mathcal{T}}^1)_{(2,1)} \\ &= \text{col}(\text{col}(0, 0, 0, 0), \text{col}(0, 0), 1), \\ \text{col}(\check{B}_0^{(2,0)}, \check{B}_1^{(2,0)}, \check{B}_{2,(0)}^{(2,0)}) &= \begin{pmatrix} \check{X}_{\mathcal{T}}^0 & \check{X}_{\mathcal{T}}^1 & (\check{X}_{\mathcal{T}}^2)_{(0)} \end{pmatrix}^{\dagger} (\check{X}_{\mathcal{T}}^2)_{(1,0)} \\ &= \text{col}(\text{col}(0, 0, 0, -2), \text{col}(0, 0, -\frac{3}{2}, -\frac{3}{2}), \text{col}(0, 0)). \end{aligned}$$

Hence, the matrices $\check{H}_0^{(j)}, \check{H}_1^{(j)}, \check{H}_{2,(0)}^{(j)}, \check{H}_2^{(j)}$ for $j = 1, 2$, in (3.12) are equal to:

$$\begin{aligned} \check{H}_0^{(1)} &= \mathbf{0}_{4 \times 2}, \quad \check{H}_1^{(1)} = \begin{pmatrix} \mathbf{0}_{4 \times 1} & \check{B}_0^{(1,1)} \end{pmatrix} = \mathbf{0}_{4 \times 2}, \\ \check{H}_{2,(0)}^{(1)} &= \begin{pmatrix} \mathbf{0}_{2 \times 1} & \check{B}_{1,(0)}^{(1,1)} \end{pmatrix} = \mathbf{0}_2, \quad \check{H}_2^{(1)} = \begin{pmatrix} 0 & -\check{B}_1^{(1,1)} \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 0 & 1 \end{pmatrix}, \\ \check{H}_0^{(2)} &= \begin{pmatrix} \check{B}_0^{(2,0)} & \mathbf{0}_{4 \times 1} \end{pmatrix} = \text{row}(\text{col}(0, 0, 0, -2), \mathbf{0}_{4 \times 1}), \end{aligned}$$

$$\begin{aligned}\check{H}_1^{(2)} &= \begin{pmatrix} \check{B}_1^{(2,0)} & \mathbf{0}_{4 \times 1} \end{pmatrix} = -\frac{3}{2} \text{row}(\text{col}(0, 0, 1, 1), \mathbf{0}_{4 \times 1}), \\ \check{H}_{2,(0)}^{(2)} &= \begin{pmatrix} \check{B}_{2,(0)}^{(2,0)} & \mathbf{0}_{2 \times 1} \end{pmatrix} = \mathbf{0}_2, \quad \check{H}_2^{(2)} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}.\end{aligned}$$

We now compute

$$\begin{aligned}\check{H}_0 &= \check{H}_0^{(1)} + \check{H}_0^{(2)} = \text{row}(\text{col}(0, 0, 0, -2), \mathbf{0}_{4 \times 1}), \\ \check{H}_1 &= \check{H}_1^{(1)} + \check{H}_1^{(2)} = -\frac{3}{2} \text{row}(\text{col}(0, 0, 1, 1), \mathbf{0}_{4 \times 1}), \\ \check{H}_{2,(0)} &= \begin{pmatrix} \check{B}_{2,(0)}^{(2,0)} & \check{B}_{1,(0)}^{(1,1)} \end{pmatrix} = \mathbf{0}_2, \quad \check{H}_2 = \begin{pmatrix} 1 & -\check{B}_1^{(1,1)} \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}, \\ H_0 &= Q\check{H}_0 = \check{H}_0 = \text{row}(\text{col}(0, 0, 0, -2), \mathbf{0}_{4 \times 1}), \\ H_1 &= Q\check{H}_1 = -\frac{3}{2} \text{row}(\text{col}(0, 0, 1, 1), \mathbf{0}_{4 \times 1}), \\ G &= \text{col}(H_0, H_1, \check{H}_{2,(0)}).\end{aligned}$$

Define $\mathcal{H}_0$, $\mathcal{H}$, and K as in (3.16):

$$\mathcal{H}_0 = \begin{pmatrix} T_0 \\ T_1 \end{pmatrix}, \quad \mathcal{H} = \begin{pmatrix} T_1 & T_2 \\ T_2 & T_3 \end{pmatrix}, \quad K = \begin{pmatrix} T_3 \\ T_4 \end{pmatrix}.$$

Note that

$$k = \text{rank}(\mathcal{H} \ K) - \text{rank} \mathcal{H} = 6 - 6 = 0.$$

Therefore, we can choose $P_1 := I_2$ and hence $P = Q(P_1 \oplus I_2) = I_4$. We decompose $\hat{K} = KP = K$ as in (3.18) (using $p = 4, p_0 = 2, m = 2, \text{card } \mathcal{A} = 1$):

$$\begin{aligned}\hat{K}_1 &= \text{col}(-10, 0, 0, 0, 18, 0, 0, 0), \\ \hat{K}_2 &= \text{col}(0, -20, -10, -10, 0, 36, 18, 18), \\ \hat{K}_3 &= \text{row}(\text{col}(0, -10, -10, -10, 0, 18, 18, 18), \text{col}(0, -10, -10, -10, 0, 18, 18, 18)).\end{aligned}$$

Computing J satisfying (3.23) by

$$J := \begin{pmatrix} \mathcal{H} & \hat{K}_1 \end{pmatrix}^\dagger \hat{K}_2 = \text{col}(0, -2, 0, 0, 0, -3, 0, 0, 0),$$

and computing

$$\begin{aligned}\hat{G} &= (I_8 \oplus P_1^T)G\check{H}_2^{-1} \\ &= \text{row} \left(\text{col}(0, 0, 0, -2, 0, 0, -\frac{3}{2}, -\frac{3}{2}, 0, 0), \text{col}(0, 0, 0, -2, 0, 0, -\frac{3}{2}, -\frac{3}{2}, 0, 0) \right),\end{aligned}$$

we have

$$\hat{K}_2 = \begin{pmatrix} \mathcal{H} & \hat{K}_1 \end{pmatrix} J \quad \text{and} \quad \hat{K}_3 = \begin{pmatrix} \mathcal{H} & \hat{K}_1 & \hat{K}_2 \end{pmatrix} \hat{G}.$$

Let $\hat{Z}_1 := (0)$, use (3.30) to compute

$$\hat{Z}_2 = (0), \quad \hat{Z}_3 = (0 \ 0), \quad \hat{Z}_4 = (-68), \quad \hat{Z}_5 = (-34 \ -34), \quad \hat{Z}_6 = \begin{pmatrix} -34 & -34 \\ -34 & -34 \end{pmatrix}$$

and define

$$\hat{Z} \equiv \begin{pmatrix} \hat{Z}_1 & \hat{Z}_2 & \hat{Z}_3 \\ \hat{Z}_2^T & \hat{Z}_4 & \hat{Z}_5 \\ \hat{Z}_3^T & \hat{Z}_5^T & \hat{Z}_6 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & -68 & -34 & -34 \\ 0 & -34 & -34 & -34 \\ 0 & -34 & -34 & -34 \end{pmatrix}.$$

We can easily check that the matrix $\widehat{Z}_1$ satisfies (3.29). Let $\widehat{T}_2 = T_2 P = T_2$. Next we compute matrices U_1 and U_2 by (3.44):

$$\begin{pmatrix} U_1 \\ U_2 \end{pmatrix} = \begin{pmatrix} \mathcal{H}_0 & \mathcal{H} \\ \widehat{T}_2 & \widehat{K}^T \end{pmatrix}^\dagger \begin{pmatrix} \widehat{K}_1 \\ \widehat{Z}_1 \\ \widehat{Z}_2^T \\ \widehat{Z}_3^T \end{pmatrix} = \text{col}(17, 0, 0, 0, 66, 0, 0, 0, 31, 0, 0, 0).$$

Write $J = \begin{pmatrix} J_1 \\ J_2 \end{pmatrix}$, where $J_1 = \text{col}(0, -2, 0, 0, 0, -3, 0, 0)$ and $J_2 = (0)$, and further decompose

$$J_1 = \begin{pmatrix} J_{1,0} \\ J_{1,1} \end{pmatrix}, \quad U_2 = \begin{pmatrix} U_{2,0} \\ U_{2,1} \end{pmatrix}, \quad \widehat{G} = \begin{pmatrix} \widehat{G}_1 \\ \widehat{G}_2 \\ \widehat{G}_3 \end{pmatrix},$$

as in (3.50), where

$$\begin{aligned} J_{1,0} &= \text{col}(0, -2, 0, 0), & J_{1,1} &= \text{col}(0, -3, 0, 0), \\ U_{2,0} &= \text{col}(66, 0, 0, 0), & U_{2,1} &= \text{col}(31, 0, 0, 0), \\ \widehat{G}_1 &= \text{row} \left(\text{col}(0, 0, 0, -2, 0, 0, -\frac{3}{2}, -\frac{3}{2}), \text{col}(0, 0, 0, -2, 0, 0, -\frac{3}{2}, -\frac{3}{2}) \right), \\ \widehat{G}_2 &= \widehat{G}_3 = (0 \ 0). \end{aligned}$$

As in (3.48), (3.51), (3.54), define

$$\begin{aligned} Q_0 &:= \begin{pmatrix} U_{2,0} & J_{1,0} & H_0 \check{H}_n^{-1} \end{pmatrix} = \begin{pmatrix} 66 & 0 \\ 0 & -2 \end{pmatrix} \oplus \begin{pmatrix} 0 & 0 \\ -2 & -2 \end{pmatrix}, \\ Q_1 &:= \begin{pmatrix} U_{2,1} & J_{1,1} & H_1 \check{H}_n^{-1} \end{pmatrix} = \begin{pmatrix} 31 & 0 \\ 0 & -3 \end{pmatrix} \oplus \begin{pmatrix} -\frac{3}{2} & -\frac{3}{2} \\ -\frac{3}{2} & -\frac{3}{2} \end{pmatrix}, \\ Q_2 &:= P \begin{pmatrix} 1 & -J_2 & -\widehat{G}_2 \\ 0 & 1 & -\widehat{G}_3 \\ \mathbf{0}_{2,1} & \mathbf{0}_{2,1} & I_2 \end{pmatrix} = I_4. \end{aligned}$$

Let $Z := P \widehat{Z} P^T = Z$, $T_5 := Z$ and $T_6 := \text{row}(T_i)_{i \in [3;5]} (M_{\mathcal{T}}(2))^\dagger \text{col}(T_i)_{i \in [3;5]}$. The matrix polynomial $H(x)$ (see (3.56)), defined by

$$H(x) = (x-1)^3 Q_2 - (x-1)^2 Q_1 - (x-1) Q_0 - (U_1 \ \mathbf{0}_{4,3})$$

is a block column relation of $M_{\mathcal{S}^{(e)}}(3)$, where $\mathcal{S}^{(e)} = (S_i)_{i \in [0;6]}$ is as in (3.55). Its determinant is the following:

$$\det H(x) = x^2(x-1)^5(x+1)^2(x^3 - 34x^2 - x + 17).$$

The atoms of the minimal representing measure μ for L are the zeroes of $\det H(x)$. Namely, writing $\mu = \sum_{j=1}^6 \delta_{x_j} A_j$, we have

$$x_1 = -1, \quad x_2 = 0, \quad x_3 = 1, \quad x_4 \approx -0.7143, \quad x_5 \approx 0.6996, \quad x_6 \approx 34.0147.$$

It remains to find the masses A_j by computing

$$\text{col}(A_i)_{i \in [6]} = \left(V_{(x_i)_{i \in [6]}}^{-1} \otimes I_4 \right) \text{col}(S_i)_{i \in [0;5]},$$

where $V_{(x_i)_{i \in [6]}} = (x_j^{i-1})_{i,j=1}^6$ is the Vandermonde matrix with i -th row equal to $\text{row}(x_j^{i-1})_{j \in [6]}$. The masses are:

$$\begin{aligned} A_1 &= \mathbf{0}_1 \oplus \begin{pmatrix} 2 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}, \quad A_2 = 2A_1, \quad A_3 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \oplus \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \\ A_4 &\approx (1.9792) \oplus \mathbf{0}_3, \quad A_5 \approx (2.0208) \oplus \mathbf{0}_3, \\ A_6 &\approx (7.4734 \cdot 10^{-7}) \oplus \mathbf{0}_3. \end{aligned}$$

We see that $x_3 = t = 1$ and $\text{rank } A_3 = 2$, whence $\text{mult}_\mu t = m = 2$.

Remark 7.1. (1) The corresponding masses A_j for the constructed measure μ are obtained via

$$(\text{col}(A_i)_{i \in [\ell]})^T = \left(V_{(x_i)_{i \in [\ell]}}^{-1} \otimes I_p \right) \text{col}(S_i)_{i \in [0; \ell-1]}$$

as in [ZZ26, Remark 3.2(3)]. If $\ell > 2n + 3$, then not all S_i are given. Computing

$$T_r = \text{row}(T_i)_{i \in [r-n-1; r-1]} \text{col}((U_1 \quad \mathbf{0}_{p \times (m - \text{card } \mathcal{A} + p - p_0)}) Q_n^{-1}, \text{col}(Q_{i-1} Q_n^{-1})_{i \in [n]})$$

for $r = 2n + 3, 2n + 4, \dots, \ell - 1$, where U_1 is as in (3.44) and Q_i are as in (3.51) and (3.54), and defining

$$\mathcal{T}^{(e_1)} := (T_i)_{i \in [0; \ell-1]},$$

we can compute the missing moments $S_{2n+3}, S_{2n+4}, \dots, S_{\ell-1}$ by

$$S_r := L_{\mathcal{T}^{(e_1)}}((x+t)^r), \quad r = 2n + 3, 2n + 4, \dots, \ell - 1.$$

(2) The matrix polynomial $W(x) := x^{n+1}I_p - \sum_{i=0}^n x^i W_i$, where

$$\text{col}(W_i)_{i \in [0; n]} := M_{\mathcal{S}}(n)^\dagger \text{col}(S_i)_{i \in [n+1; 2n+1]}$$

is also a block column relation of $M_{\mathcal{S}^{(e)}}(n+1)$ and $\mathcal{S}^{(e)}$ is as in (3.55). However, it is not clear from the factorization of the determinant of $W(x)$, which zeroes of $\det W(x)$ are the atoms of the measure μ . For example, using the data given in the example above, we have

$$\text{col}(W_i)_{i \in [0; 2]} := M_{\mathcal{S}}(2)^\dagger \text{col}(S_i)_{i \in [3; 5]},$$

which gives us

$$W_0 = \begin{pmatrix} -17 & 0 \\ 0 & 0 \end{pmatrix} \oplus \begin{pmatrix} \frac{1}{5} & 0 \\ -\frac{1}{5} & 0 \end{pmatrix}, \quad W_1 = I_2 \oplus \begin{pmatrix} \frac{13}{20} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} \end{pmatrix}, \quad W_2 = \begin{pmatrix} 34 & 0 \\ 0 & 0 \end{pmatrix} \oplus \begin{pmatrix} \frac{3}{20} & -\frac{1}{4} \\ -\frac{1}{4} & -\frac{1}{4} \end{pmatrix}.$$

Defining the matrix polynomial

$$W(x) = x^3 I_4 - x^2 W_2 - x W_1 - W_0,$$

we have

$$\det W(x) = \frac{1}{10} x^2 (x-1)^2 (x+1)^2 (x^3 - 34x^2 - x + 17) (10x^3 + x^2 - 1).$$

We can see that all of the atoms of the measure μ constructed in the example are the zeroes of $\det W(x)$, but just using the matrix polynomial $W(x)$ to find the atoms of the representing measure for the given linear operator, we have no direct way of knowing that the zeroes of the factor $10x^3 + x^2 - 1$ are not the atoms of the measure.

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